

FARMERCONNECT PLATFORM

Complete Product Requirements Document — Billion-Dollar Edition

Version 1.0 | Confidential & Proprietary

Architecture: Zero-INR Infrastructure · Serverless-First · Globally Scalable

Vision Statement: Build the most trusted, inclusive, and intelligent property + agricultural land platform in India — eliminating brokers, reducing friction, and connecting every property type (urban residential, rural agricultural, commercial, PG/co-living) directly between owners and seekers — powered by AI, secured at bank grade, and operated at zero infrastructure cost.

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SECTION 1 — EXECUTIVE SUMMARY

1.1 Product Identity

Field	Detail
Product Name	FarmerConnect
Tagline	<i>Every Property. Every Acre. Zero Broker.</i>
Category	PropTech + AgriTech Marketplace
Geography	India (Phase 1: Tier 1-2 cities + rural Andhra Pradesh, Telangana, Karnataka)
Target TAM	₹4,50,000 Cr+ Indian real estate + ₹12,000 Cr agri-land market
Launch Target	10,000 listings · 50,000 users in first 90 days
Infrastructure Cost	₹0/month (zero-INR serverless stack)
Maintenance Model	Self-healing, auto-scaling, event-driven

1.2 The Problem We Solve

India's property market is deeply broken across five dimensions:

1. **Urban Renters** pay 1-2 months' brokerage (₹10,000–₹60,000) for every house move — often every 11 months due to short lease cycles.

2. **Agricultural Landowners** have NO digital platform to list, lease, or sell farm land — relying entirely on local middlemen who charge 3–5% commissions.

3. **Rural Migrants** moving to cities for work have zero trusted platform to find affordable PG or shared accommodation in safe localities.

4. **Property Owners (NRI & Local)** struggle to find verified tenants and manage rent collection, agreements, and maintenance remotely.

5. **Society Admins** have fragmented tools for visitor management, complaint tracking, payments collection, and community communication.

1.3 The FarmerConnect Solution

A unified, AI-first, zero-broker property marketplace that serves:

- Urban rental seekers
- Agricultural land seekers and owners
- PG/co-living seekers
- Society communities
- Property managers and NRI landlords

What makes it a ₹1,000 Cr+ platform:

- Uniquely covers agricultural + urban properties (no competitor does both)
- Zero-broker direct connect with AI-powered matching
- Built on ₹0/month infrastructure (Supabase + Cloudflare + Vercel)
- Bank-grade security without enterprise cost
- Vernacular-first (Telugu, Kannada, Hindi, Tamil alongside English)

SECTION 2 — MARKET OPPORTUNITY & COMPETITIVE ANALYSIS

2.1 Market Size Breakdown

Segment	TAM (India)	FarmerConnect Claim (Yr 3)
Urban Rental Market	₹1,80,000 Cr/year	0.8% = ₹1,440 Cr
Agricultural Land Lease/Sale	₹12,000 Cr/year	3% = ₹360 Cr
PG/Co-living	₹1,20,000 Cr/year	0.5% = ₹600 Cr
Society Management SaaS	₹8,500 Cr/year	2% = ₹170 Cr
Ancillary (Agreements, Movers)	₹25,000 Cr/year	1% = ₹250 Cr
Total Addressable	₹3,45,500 Cr	₹2,820 Cr

2.2 Competitive Positioning Matrix

Platform	Urban Rental	Agri Land	PG/Co-living	Society Mgmt	Zero Broker	Vernacular
NoBroker	✓	✗	✗	✗	✓	✗
MagicBricks	✓	✗	✗	✗	✗	✗
99acres	✓	✗	✗	✗	✗	✗
Housing.com	✓	✗	✗	✗	✗	✗
Zolo/Stanza	✗	✗	✓	✗	✓	✗
MyGate	✗	✗	✗	✓	N/A	✗
FarmerConnect	✓	✓	✓	✓	✓	✓

FarmerConnect is the **ONLY** platform covering all six dimensions simultaneously.

2.3 Unique Differentiators (Moat Builders)

1. **Agricultural Land Module** — First-ever structured digital marketplace for farm land (lease, sale, joint farming arrangements)
2. **Vernacular-First Interface** — Full Telugu, Kannada, Hindi support with voice search
3. **Offline-Capable PWA** — Works in low-connectivity rural areas via service workers
4. **AI Matchmaker** — Proprietary scoring engine that matches properties to seekers based on stated + inferred preferences
5. **Zero Broker Guarantee** — Platform-level mechanism that detects and penalizes broker infiltration
6. **Instant Digital Agreement** — E-sign rental + land lease agreements in under 10 minutes
7. **Transparent Price Intelligence** — AI-predicted fair market rent based on locality, BHK, season, and supply-demand

SECTION 3 — PRODUCT VISION, MISSION & PRINCIPLES

3.1 Vision

"To democratize property access for every Indian — from a farm worker in Nellore to a software engineer in Bengaluru — through trust, transparency, and technology."

3.2 Mission

Build a platform so trustworthy and simple that a farmer in rural Andhra Pradesh can list his 5-acre land for lease with the same confidence as a Bengaluru techie finding a 2BHK — in their own language, from their own phone.

3.3 Core Design Principles

Principle	Meaning in Practice
Trust First	Every user, listing, and transaction is verified before going live
Zero Friction	Any key action (search, contact, pay) completes in ≤3 taps
Privacy by Default	Phone numbers never exposed; calls go through masked numbers
Vernacular Native	No user should be disadvantaged by their language
Offline Resilient	Core search and viewing works without internet
Transparent Pricing	No hidden fees; all charges shown upfront before confirmation
Data Sovereignty	User data never sold; DPDP Act 2023 compliant by design

SECTION 4 — BUSINESS MODEL & REVENUE ARCHITECTURE

4.1 Revenue Streams

Stream 1: Subscription Plans (Primary)

For Property Seekers (Tenants/Buyers):

Plan	Price	Duration	Features
Free	₹0	Always	Browse, shortlist, 3 contacts/month
Connect	₹499	30 days	Unlimited contacts, chat, visit scheduling
Relax	₹1,499	90 days	RM-assisted search, priority matching, home visits arranged
Power	₹2,999	180 days	All of Relax + legal agreement included, background check of owner

For Property Owners/Landlords:

Plan	Price	Duration	Features
Free	₹0	Always	1 active listing, basic photos
Premium	₹799	90 days	5 listings, featured placement, unlimited leads
Pro	₹1,999	180 days	15 listings, top of search, RM support, verified badge
Elite	₹4,999	365 days	Unlimited listings, dedicated RM, monthly reports, NRI support

For Agricultural Land Owners:

Plan	Price	Duration	Features
Basic	₹0	Always	1 land listing, basic details
AgriPro	₹1,499	180 days	5 listings, verified badge, priority in agri search
AgriElite	₹3,999	365 days	Unlimited listings, legal support, yield data integration

Stream 2: Transaction Fees (Secondary)

Service	Fee	Trigger
Digital Rent Agreement	₹399–₹2,499	User initiates agreement
Digital Land Lease Agreement	₹599–₹3,499	User initiates agri agreement
E-stamp (pass-through)	State stamp duty	Agreement execution
Rent Payment Processing	1.5% of rent (capped ₹999)	Successful rent payment

Service	Fee	Trigger
Background Verification	₹199-₹599	KYC verification request
Packers & Movers Referral	5-8% commission	Completed booking

Stream 3: Advertising & Promotions (Tertiary)

Product	Description	Price Range
Featured Listing	Top placement in search results	₹299-₹999/listing/week
Homepage Banner	Property developer banners	₹10,000-₹50,000/week
Locality Spotlight	Builder pins on map	₹5,000-₹20,000/month
Push Notification Ads	Targeted push to seekers	₹2/notification

Stream 4: B2B SaaS (Society Management)

Module	Fee	Billing
Society Management Suite	₹50-₹150/unit/month	Monthly
Visitor Management	₹30/unit/month	Monthly
Maintenance Collection	1% of collection	Per transaction
Community Commerce	8% commission	Per order

4.2 Revenue Projection (Conservative)

Year	Users	Paying Users (5%)	ARPU (₹/year)	Total Revenue
Yr 1	1,00,000	5,000	₹1,800	₹90 Lakhs
Yr 2	10,00,000	75,000	₹2,400	₹18 Crores
Yr 3	50,00,000	5,00,000	₹3,200	₹160 Crores
Yr 5	2,00,00,000	30,00,000	₹4,000	₹1,200 Crores

4.3 Unit Economics

- **CAC (Customer Acquisition Cost):** ₹45–₹80 (organic SEO + referral-first)
 - **LTV (Lifetime Value):** ₹3,200–₹8,400 (3–7 year average tenure)
 - **LTV:CAC Ratio Target:** 40:1
 - **Gross Margin:** 82–88% (software-only, minimal ops)
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SECTION 5 — USER TYPES, PERSONAS & ROLE MATRIX

5.1 External User Personas

Persona 1: Ravi — The Urban Tenant

- **Age:** 26, Software Engineer, Bengaluru
- **Pain:** Pays ₹25,000 broker fee every time he moves; doesn't trust listings
- **Goal:** Find a verified 1BHK near office within ₹18,000/month, no broker
- **Device:** Android (OnePlus), high-speed internet
- **Language:** English + Kannada

Persona 2: Lakshmi — The Owner (NRI)

- **Age:** 52, Working in UAE, owns 2 flats in Hyderabad
- **Pain:** Can't trust local brokers; rent collected irregularly; agreements not digitized
- **Goal:** Verified tenants, online rent collection, instant agreement, monthly reports
- **Device:** iPhone, WhatsApp-primary
- **Language:** Telugu + English

Persona 3: Ramaiah — The Farmer

- **Age:** 58, Owns 8 acres in Krishna district, AP
- **Pain:** Wants to lease land to a reliable farmer but has no trusted platform; local dalals take 3% cut
- **Goal:** Find a verified lessee for his land at fair market price
- **Device:** Android (Samsung), 4G connection
- **Language:** Telugu only

Persona 4: Priya — The PG Seeker

- **Age:** 21, College Student, new to Hyderabad
- **Pain:** Unknown PG quality, safety concerns, hidden costs
- **Goal:** Find a safe, affordable, verified PG near college with meals
- **Device:** iPhone, Instagram-native
- **Language:** Hindi + Telugu

Persona 5: Suresh — The Society Admin

- **Age:** 44, Retired Colonel, RWA Secretary
- **Pain:** Manual collection of maintenance fees, paper complaints, no visitor log
- **Goal:** Digital society management: payments, complaints, visitors, notices
- **Device:** iPad + Android phone
- **Language:** English

Persona 6: Kiran — The Broker (Power User)

- **Age:** 35, Real estate broker, 12 years experience
- **Pain:** Digital tools are expensive; no CRM for managing 200+ clients
- **Goal:** List inventory, manage leads, schedule visits, earn commission transparently
- **Device:** Android, WhatsApp-heavy
- **Language:** Telugu + Hindi + English

5.2 Internal Roles Matrix

Role	Access Level	Key Responsibilities
Relationship Manager (RM)	L2	Manage tenant/owner plans; schedule visits; confirm deal closure
Verification Agent	L2	KYC review; document verification; listing approval
Agreement Executive	L3	Draft, stamp, dispatch rental/land agreements
Customer Support	L1	Handle tickets; escalate disputes; refund processing
Finance Admin	L4	Settlement reports; reconciliation; fraud flags

Role	Access Level	Key Responsibilities
Product Admin	L5	Feature flags; user management; system configs
Super Admin	L6	Full platform access; compliance exports; audit logs
AI/ML Ops	L4	Model monitoring; bias audits; data pipeline management

5.3 Permission Matrix

Action	Tenant	Owner	Broker	Society Admin	RM	Verifier	Finance	Super Admin
Search listings	✓	✓	✓	✓	✓	✓	✗	✓
Create listing	✗	✓	✓	✓	✗	✗	✗	✓
Contact owner	✓	✗	✓	✗	✓	✗	✗	✓
Verify KYC	✗	✗	✗	✗	✗	✓	✗	✓
Approve listing	✗	✗	✗	✗	✗	✓	✗	✓
Process payment	✓	✓	✓	✓	✗	✗	✓	✓
View all users	✗	✗	✗	✗	✗	✗	✗	✓
Export data	✗	✗	✗	✗	✗	✗	✓	✓
Override deals	✗	✗	✗	✗	✓	✗	✗	✓

SECTION 6 — CORE FEATURE MODULES — DETAILED SPECIFICATIONS

Module 1: User Onboarding & KYC

1.1 Registration Flow

Step 1: Enter phone number → OTP (Supabase Auth / MSG91)

Step 2: Select role (Tenant / Owner / Farmer / PG Seeker / Society Admin / Broker)

Step 3: Basic profile (Name, profile photo, city)
Step 4: Role-specific questions
Step 5: KYC initiation (optional at registration, mandatory for posting/paying)

1.2 KYC Data Collected by Role

Role	Required Documents
Tenant	Aadhaar (masked), Selfie liveness check
Owner	Aadhaar + PAN + Property document (sale deed / khata / EC)
Farmer	Aadhaar + Patta/ROR (Record of Rights) + Land passbook
Broker	Aadhaar + PAN + RERA registration number
Society Admin	Aadhaar + Society registration document + Election proof

1.3 KYC Verification Process

1. User uploads documents via in-app camera (no third-party redirect)
2. System runs OCR extraction (Tesseract / AWS Textract free tier)
3. Automated face match: selfie vs Aadhaar photo (DigiLocker integration)
4. Automated RERA validation (RERA API for brokers)
5. AI risk score assigned (0-100): documents quality, match confidence, fraud signals
6. Score > 80: Auto-approved
7. Score 50-80: Manual review by verification agent (SLA: 4 hours)
8. Score < 50: Rejected with reason + re-upload option
9. KYC status: Pending → In Review → Verified / Rejected
10. Verified badge shown on profile and listings

1.4 Notification on KYC

- SMS + WhatsApp + In-app push on status change
- Rejection includes specific reason and document re-upload link
- Verified owners receive automated listing approval fast-track

Module 2: Property Search & Discovery

2.1 Search Architecture

Input Layer:

- Text search (property name, locality, landmark)
- Voice search (vernacular: Telugu, Hindi, Kannada, Tamil)
- Map-based drag-to-search
- AI-assisted: "2BHK near Miyapur metro under ₹15,000"

Processing Layer (Supabase + pgvector):

- Full-text search index on listings (tsvector)
- Vector embeddings for semantic similarity search
- Geospatial PostGIS queries for radius/polygon search
- Real-time inventory check (Supabase Realtime)

Results Layer:

- Relevance ranking: AI score × recency × completeness × verification
- Personalized ranking: user preference model
- Sponsored insertion: 1 in 10 positions

2.2 Filter Taxonomy

For Residential Rental:

- Property type: Flat, Independent House, Villa, Row House
- BHK: 1RK, 1BHK, 2BHK, 3BHK, 4BHK, 4BHK+
- Rent range: Slider (₹3,000 – ₹5,00,000)
- Furnishing: Unfurnished, Semi-Furnished, Fully Furnished
- Floor: Ground, Low (1-4), Mid (5-10), High (10+), Penthouse
- Facing: East, North, West, South
- Available from: Date picker
- Pet friendly: Yes/No/Negotiable
- Parking: 2W / 4W / Both / None
- Lift: Yes/No
- Society: Gated / Non-Gated
- Verified only: Toggle
- Owner-direct only: Toggle

For Agricultural Land:

- Land type: Agricultural, Orchard, Plantation, Fallow, Converted
- Purpose: Lease, Sale, Joint Farming, Mortgage
- Area: Slider (0.5 acres – 500 acres)
- Soil type: Red, Black (Cotton), Alluvial, Sandy, Laterite
- Water source: Borewell, Canal, Drip, Rain-fed, River
- District / Mandal / Village (cascading dropdown)
- Survey number (text search)
- Price: Per acre per year (lease) or per acre (sale)
- Crop suitability: Paddy, Cotton, Chilli, Mango, Sugarcane, Groundnut, Tomato

For PG/Co-living:

- Gender preference: Boys / Girls / Mixed
- Room type: Single / Double / Triple / Dormitory
- Meal plan: With meals / Without meals / Optional
- Amenities: WiFi, AC, Laundry, TV, Gym, Swimming pool
- College/company nearby (tag-based)
- Monthly fee range
- Deposit amount
- Notice period: 30 days / 15 days / None

2.3 Search Result Card (Data Fields)

json

```
{
  "listingId": "FC-2024-BLR-10234",
  "title": "2BHK Fully Furnished in Whitefield",
  "propertyType": "flat",
  "bhk": 2,
  "rent": 25000,
  "deposit": 75000,
  "area": 1100,
  "areaUnit": "sqft",
  "locality": "Whitefield",
  "city": "Bengaluru",
  "latitude": 12.9698,
  "longitude": 77.7499,
  "photos": ["url1", "url2", "url3"],
  "ownerVerified": true,
  "listingVerified": true,
  "postedAt": "2024-01-15T10:30:00Z",
  "availableFrom": "2024-02-01",
  "furnishing": "fully_furnished",
  "facing": "east",
  "floor": 4,
  "totalFloors": 12,
  "parking": ["2W", "4W"],
  "petFriendly": true,
  "nearbyMetro": "Whitefield Metro (0.8 km)",
  "aiMatchScore": 94,
  "views": 234,
  "contactsUnlocked": 8,
  "isPremium": true,
  "isSponsored": false
}
```

Module 3: Listing Management

3.1 Create Listing Flow (Owner)

Step 1: Property Basics

- Property type (Flat / House / Villa / Land / PG / Commercial)
- Transaction type (Rent / Sale / Lease)
- BHK selection
- City → Locality (autocomplete with MapBox)
- Society/Building name (optional autocomplete)
- Floor & Total floors
- Built-up area & Carpet area

Step 2: Photos & Media Upload

- Minimum 5 photos required
- Maximum 50 photos
- Auto-enhancement: brightness, sharpness correction via Cloudflare Images
- Mandatory: Living room, bedroom, bathroom, kitchen, exterior
- Optional: Floor plan upload (PDF/Image)
- 360° virtual tour embed (Matterport link, optional)
- Video tour (up to 3 min, compressed via Cloudflare Stream)
- AI auto-caption generation for each photo

Step 3: Property Details

- Furnishing status (unfurnished / semi / fully)
- Amenities checklist (50+ options with icons)
- Facing direction
- Age of property (slider)
- Maintenance fee (per month)
- Available from (date)
- Parking details
- Society features (gym, pool, clubhouse, security, power backup)

Step 4: Pricing

- Monthly rent (with market benchmark shown)
- Security deposit (with legal max shown: 2-3 months)
- Brokerage: ZERO (platform-enforced for owner listings)
- Any other charges (Society maintenance, water)
- AI suggested price: "Based on 47 similar properties, ₹22,000-₹26,000"

Step 5: Preferences

- Who can contact: Anyone / Verified Only / Families Only / Bachelors / Working Professionals
- Pet policy
- Smoking policy
- Religion preference: HIDDEN from seekers (for compliance); used only for algorithmic soft-preference
- Preferred tenant occupation

Step 6: Preview & Publish

- Full listing preview in seeker view
- Completeness score shown: "87% complete - Add 2 more photos for Premium badge"
- Estimated visibility score
- Publish → goes to Verification queue
- Expected approval time shown (auto-approve if owner is KYC-verified Level 2+)

3.2 Listing Verification Process

Auto-checks (instant):

- ✓ Photo clarity (AI model: blur detection, inappropriate content)
- ✓ Rent reasonableness (vs. locality benchmark ± 3 SD)
- ✓ Description spam check (no phone numbers, no "WhatsApp", no broker language)
- ✓ Duplicate listing detection (cosine similarity on address + rent + photos)
- ✓ Owner KYC status

Manual checks (verification agent, SLA 2 hours):

- ✓ Property ownership document cross-check
- ✓ Society name / building verification (Google Maps API)
- ✓ Photo relevance (is this actually the property?)
- ✓ Pricing abuse check (far below market = potential scam)

Post-verification:

- Approved: Goes live + owner notified
- Rejected: Reason given + edit option
- Suspended: For repeated violations (account-level flag)

3.3 Listing States & Lifecycle

Draft → Under Review → Active → Shortlisted (by seekers) →
Contact Unlocked → Visit Scheduled → Negotiating →
Deal Confirmed / Relisted / Expired

3.4 Listing Expiry & Refresh

- Standard listings expire after 90 days
 - System sends Day-75, Day-85, Day-89 reminders
 - One-tap refresh (re-verification if price/details unchanged)
 - Auto-expire prevents stale inventory pollution
-

Module 4: Communication (Chat, Calls, Number Masking)

4.1 Chat System

Architecture: Supabase Realtime (WebSocket)
Storage: Supabase PostgreSQL (messages table)
File sharing: Cloudflare R2 (photos, documents)
Encryption: End-to-end encrypted via Signal Protocol (libsodium)

Features:

- One-to-one chat between seeker and owner
- Typing indicators, read receipts
- Message reactions (shorthand responses)
- Photo/document sharing
- Quick replies: "Is property still available?" / "Can I visit tomorrow?" / "What is maintenance?"
- Auto-translate (vernacular ↔ English via LibreTranslate)
- AI-assisted response suggestions for owners
- Spam detection: block if > 5 unsolicited contacts/day
- Report & Block functionality
- Chat archival after deal confirmed or 90 days

4.2 Number Masking & Call System

Architecture: Exotel (free starter) / Twilio / MSG91 (pay-as-you-go)
Zero fixed cost: pay only per call minute (~₹0.25/min)

Flow:

1. Seeker clicks "Call Owner"
2. System allocates temp virtual number (masked)
3. Seeker's phone dials virtual number
4. System bridges call to owner's real number
5. Neither party sees the other's real number

6. Call recorded (stored 90 days, accessible for disputes)
7. Call log stored: duration, time, listing, outcome

Alternatives (Zero Cost):

- In-app VoIP (WebRTC via Agora free tier: 10,000 min/month free)
- WhatsApp Business API trigger (for ₹0 on template messages)

4.3 Communication Rules (Platform Policy)

- Phone numbers NEVER shared in chat or listing description (auto-redacted by AI)
- Minimum 1 KYC tier (phone-verified) required to initiate contact
- Contact unlock costs one "credit" (included in paid plans or ₹9/credit free tier)
- Anti-spam: ML model detects broker solicitation language → auto-flag

Module 5: Payments & Financial Infrastructure

5.1 Payment Architecture (Zero-INR Stack)

Primary Gateway: Razorpay (pay-per-transaction, 2% + GST)

Backup Gateway: Cashfree Payments (pay-per-transaction)

UPI: Native UPI via both gateways

Cards: Visa, Mastercard, Rupay

Netbanking: All major banks

Wallets: Paytm, PhonePe (via gateway)

EMI: Bajaj, HDFC, ICICI (for large deposits)

BNPL: LazyPay, ZestMoney

Cost Model:

- Platform absorbs 0% of transaction fees for subscription payments
- Transaction fee passed to user (transparently shown): 2% for cards/NB, 0% for UPI
- UPI preferred (zero cost) – default payment method

5.2 Rent Payment Flow

1. Tenant adds property (owner bank details or UPI)
2. System validates owner bank via penny-drop (₹1 test transfer)
3. Tenant selects amount, chooses payment method
4. Payment captured by Razorpay
5. After T+2 (settlement cycle): transferred to owner bank

6. Tenant receives digital receipt (PDF, auto-generated)
7. Owner receives payment notification (SMS + WhatsApp + in-app)
8. System logs: payer, payee, amount, property, date, reference
9. Monthly reminder auto-scheduled (customizable date)
10. Late payment flag after 5 days overdue

5.3 Subscription Payment Flow

1. User selects plan
2. Total breakup shown: Plan fee + GST (18%) + Platform fee (0%)
3. Payment method selection
4. UPI: QR or UPI ID (instant, zero cost for platform)
5. Card: Razorpay checkout (PCI DSS compliant, data stored at Razorpay)
6. On success: Plan activated instantly
7. Invoice auto-generated (GST compliant, with HSN code)
8. Renewal reminder: Day 7, Day 3, Day 1 before expiry
9. Auto-renewal: Optional (with explicit consent – DPDP compliant)

5.4 Agreement Fee Flow

1. User selects agreement type (rental / land lease)
2. System calculates: Stamp duty + FarmerConnect fee + e-stamping fee
3. Full fee breakdown shown before confirmation
4. Payment → triggers agreement drafting workflow
5. Agreement ops executes within 2 hours (Mon-Sat)
6. Digital agreement delivered to both parties
7. Physical delivery option: Speed Post (optional, ₹299 extra)

5.5 Wallet & Credits System

FarmerConnect Credits (FC Credits):

- 1 FC Credit = ₹1 purchasing power within platform
- Earned via: referrals (₹200/successful referral), reviews, milestones
- Used for: contact unlocks (₹9/contact), boost listings, premium search filters
- Expiry: 365 days from earning
- Non-withdrawable (not real money)
- GST not applicable on credit usage (pre-taxed at purchase)

Module 6: Agricultural Land Module (Unique Module)

6.1 Land Listing Structure

json

```
{
  "landId": "FC-AGR-2024-AP-5623",
  "ownerName": "Ramaiah Reddy",
  "ownerVerified": true,
  "state": "Andhra Pradesh",
  "district": "Krishna",
  "mandal": "Nandigama",
  "village": "Palavalasa",
  "surveyNumber": "123/2A",
  "area": 4.5,
  "areaUnit": "acres",
  "pattaNumber": "AP-KRI-2023-00456",
  "transactionType": "lease",
  "leaseTermYears": 3,
  "leasePricePerAcrePerYear": 18000,
  "landType": "agricultural",
  "soilType": "black",
  "waterSource": ["borewell", "canal"],
  "currentCrop": "cotton",
  "suitableCrops": ["cotton", "chilli", "sunflower"],
  "electricityConnection": true,
  "fenceStatus": "fully_fenced",
  "roadAccess": "asphalt",
  "nearestTown": "Nandigama",
  "nearestTownDistanceKm": 12,
  "documents": {
    "patta": "verified",
    "rdr": "verified",
    "ecCertificate": "verified"
  },
  "photos": ["aerial_photo", "boundary_photo", "water_source_photo"],
  "postedAt": "2024-01-20",
  "availableFrom": "2024-06-01"
}
```

6.2 Land Verification Process

Additional checks for land listings (beyond residential):

1. Survey number validation (AP/TS land records API integration)
2. Encumbrance Certificate (EC) check (automated via AP Registration API)
3. Patta/ROR cross-verification
4. Pending litigation check (court records API, best-effort)
5. Land measurement validation (satellite image area estimation via MapBox)
6. Owner name match: Aadhaar → Patta (same name)

Verification Badge Levels:

- Bronze: Phone + Aadhaar verified
- Silver: + Document verified (Patta + EC)
- Gold: + Physical inspection by FC field agent (Phase 2)

6.3 Land Lease Agreement Generation

Template: Customizable by state (AP/Telangana/Karnataka specific templates)

Parties: Owner (lessor) + Lessee (tenant farmer)

Content:

- Survey number, area, village, district
- Lease period, start date, end date
- Annual/seasonal lease amount
- Payment schedule (pre-harvest / post-harvest / quarterly)
- Water rights specification
- Crop restriction clauses (if any)
- Renewal terms
- Dispute resolution: Lok Adalat / Arbitration

E-stamp: AP/TS stamp paper (via eStamp portal integration)

E-sign: Aadhaar-based e-sign (DigiLocker / NSDL eSigning)

Delivery: PDF + registered post (optional)

6.4 Crop Price Integration

Source: Agmarknet API (Government of India, free)

Display: Current MSP (Minimum Support Price) for crops

Use case: Lessee sees current crop prices → makes informed leasing decision

Alert: Owner notified if nearby market price rises (lease renegotiation signal)

Module 7: PG / Co-Living Module

7.1 PG Listing Fields (Additional)

- PG Name, owner name, RERA registration (optional)
- Total rooms: Single / Double / Triple / Dorm
- Gender preference: Boys / Girls / Any
- Total beds available vs. occupied
- Meal plan: (None / Breakfast / All meals)
- Meal timing, cuisine, cuisine preferences
- WiFi speed (Mbps), provider
- Common areas: TV room, study room, gym, terrace
- Security: CCTV, biometric entry, security guard
- Caretaker: Resident / On-call
- Notice period: 15/30/60 days
- Additional rules (no late entry, no guests, alcohol policy)
- Monthly all-inclusive price (clarify what's included)
- One-time joining fee
- Deposit amount

7.2 PG Discovery Features

- College-proximity search: Filter by college name (drop-down from DB)
- Company-proximity search: Filter by company/IT park name
- Automated commute time estimation (MapBox Directions API)
- Aggregated PG reviews (Google-style: food, cleanliness, safety, owner behavior)
- Real-time bed availability indicator
- Waitlist feature for full PGs

7.3 PG Booking Flow

1. Seeker finds PG → views photos, reviews, amenities
2. "Check Availability" → real-time bed status
3. Schedule visit → owner/caretaker confirms time slot
4. Visit confirmation: QR code to show at gate
5. Post-visit: Offer to book → pay ₹1,000 token (refundable if cancelled 48h before move-in)
6. Move-in: Photo documentation (checklist: room condition, furniture inventory)
7. Monthly rent auto-reminder set
8. Review request sent after 30 days

Module 8: Society Management Module

8.1 Society Admin Onboarding

1. Admin creates society profile (name, address, RWA registration, unit count)
2. Verification: RWA registration document + admin election proof
3. Invite residents: Upload CSV of flat numbers + owner names → auto-generate invite links
4. Resident app setup: Resident claims their flat via OTP-verified invite

8.2 Society Features

Maintenance Billing & Collection:

- Admin sets maintenance amount per unit (or per sqft)
- Auto-invoice generated on 1st of each month
- Resident receives bill via app + WhatsApp
- Payment options: UPI, card, netbanking
- Real-time collection dashboard for admin
- Penalty: Auto-calculated for late payments (configurable %)
- Defaulter report: Downloadable CSV
- Bank account: Society's registered account (after penny-drop verification)

Visitor Management:

- Resident pre-approves visitor via app (generates QR code)
- Guard scans QR at gate → instant approve
- Unannounced visitor: Guard enters name → resident notified → 1-tap approve/deny
- Delivery: One-tap delivery approval (DHL, Amazon, Zomato templates)
- Cab entry: Auto-approve via OTP-linked cab number
- Visitor log: Searchable, 180-day retention

Community Noticeboard:

- Admin posts notices (PDF or text)
- Residents post buy/sell/borrow requests
- Emergency alerts: Broadcast with push + SMS
- Polls: Admin creates → residents vote → results public
- Events: Create event → RSVP → reminder

Maintenance Requests:

- Resident raises ticket (category: plumbing, electrical, lift, common area)
- Photo attachment
- Assigned to maintenance staff by admin
- Status: Open → Assigned → In Progress → Resolved → Rated
- SLA tracking: Admin sees overdue tickets
- Vendor management: Add trusted vendors to society panel

Service Partners (Revenue):

- Society recommends: Housekeeping, tiffin, water cans, pest control
- Vendor listed in app: "Society-approved vendors"
- Resident orders from partner → FarmerConnect earns 8% commission
- Vendor performance: Rated by residents

Module 9: Support, Ticketing & RM Dashboard

9.1 Support Ticket System

Channels:

- In-app chat (Chatwoot, self-hosted on Railway free tier)
- WhatsApp Business API (incoming messages routed to Chatwoot)
- Email (support@farmerconnect.in → Chatwoot)

Ticket Categories:

- Listing issue (fake, wrong info, scam concern)
- Payment issue (failed, not credited, refund)
- Agreement issue (delay, error)
- Account issue (login, KYC, plan activation)
- Deal dispute (security deposit, property condition)
- Other

SLA:

- L1 (Support): 4 hours (Mon-Sat), 24 hours (Sun)
- L2 (RM/Finance): 24 hours
- L3 (Legal/Compliance): 72 hours

Auto-resolution:

- Duplicate tickets: Auto-merged
- Known issues: Auto-response with solution
- Payment confirmation queries: Auto-resolved via Razorpay webhook

9.2 RM Dashboard

RM sees:

- Assigned tenants (Relax plan holders)
- Tenant preference card (location, budget, BHK, move-in date)
- Shortlisted properties (by system + manually curated)
- Scheduled visits (calendar view)
- Deal status: Open / Visiting / Negotiating / Closed / Lost

RM actions:

- Add property to tenant shortlist
- Schedule visit (owner + tenant notified)
- Mark visit complete + add notes
- Initiate deal confirmation
- Request agreement from agreement ops
- Mark deal "closed" (triggers payout/commission report)

RM performance KPIs:

- Closures per month
- Avg. time to close (days)
- NPS score from assigned tenants
- Tenant plan renewal rate

SECTION 7 — USER JOURNEYS — STEP-BY-STEP SCREEN FLOWS

Journey 1: Tenant Finds and Rents a Property

[DISCOVERY PHASE]

Screen 1: Home → "I'm Looking for a Property" CTA

Screen 2: Search bar + voice input + "Near Me" quick filter

Screen 3: Results grid (10 cards per load, infinite scroll)

Each card: photo, BHK, rent, locality, verified badge, AI match %

Screen 4: Map view toggle → properties shown as pins

Screen 5: Open listing → full gallery, all details, AI summary ("Perfect for 2 working professionals")

[SHORTLISTING PHASE]

Screen 6: Heart icon = shortlist (saved to profile)

Screen 7: Compare mode: Side-by-side 2 properties

Screen 8: "Share" → WhatsApp share with property card

[CONTACT PHASE]

Screen 9: "Contact Owner" → Plan check → if free plan: "Upgrade for unlimited" or "Use 1 credit"

Screen 10: Chat initiated → pre-filled: "Hi, I'm interested in [property]. Is it available?"

Screen 11: Call option (number-masked WebRTC call)

Screen 12: Schedule visit → date picker → time slot → confirmation

[DEAL PHASE]

Screen 13: Visit completed → tenant marks "Interested"

Screen 14: "Proceed to Agreement" → enter terms (rent, deposit, move-in date)

Screen 15: Agreement form auto-filled → review → e-sign (Aadhaar OTP)

Screen 16: Pay stamp duty + agreement fee

Screen 17: Agreement generated → both parties receive PDF

Screen 18: Set up rent payment → auto-reminder configured

Screen 19: Deal marked "Confirmed" → both parties get confirmation notification

[POST-MOVE PHASE]

Screen 20: Rate owner and property (post 7 days)

Screen 21: Monthly rent payment reminder → one-tap pay

Screen 22: Support channel for any issues

Journey 2: Owner Posts a Listing

Screen 1: "Post Property" CTA (authenticated user)

Screen 2: Property type selection (grid of icons)

Screen 3: Map → drop pin to mark location → address auto-filled

Screen 4: Basic details form (BHK, floor, area, facing)

Screen 5: Photo upload → guidance overlay ("Take a bright, well-framed photo")

Screen 6: Amenities checklist (50 items with icons, multi-select)

Screen 7: Rent input → AI benchmark shown: "Similar properties: ₹18,000-₹22,000"

Screen 8: Tenant preferences (family/bachelor/any)

Screen 9: Preview full listing

Screen 10: Publish → verification queue → "Under Review (Est. 2 hours)"

Screen 11: Notification: "Listing Live!" → share link

Screen 12: Owner dashboard: Views, contacts, messages

Screen 13: Lead management: Interested seekers list

Screen 14: Visit scheduler: Accept/decline visit requests

Screen 15: Deal confirmation: Mark as "Rented"

Journey 3: Farmer Lists Agricultural Land

Screen 1: Home → "I'm a Farmer/Land Owner" → "List My Land"

Screen 2: Land type selection (Agricultural / Orchard / Vacant / Mixed)

Screen 3: Survey number + District/Mandal/Village cascading

Screen 4: Area input + unit (acres / cents / guntas – regional units)

Screen 5: Document upload (Patta, ROR) → OCR auto-fills details

Screen 6: Water source, soil type, current crop, suitable crops

Screen 7: Transaction type: Lease (with duration) or Sale

Screen 8: Price input → per acre per year (lease) or per acre (sale)

Screen 9: Land photos (aerial view recommended via drone or satellite capture)

Screen 10: Verification queue → Land records API cross-check

Screen 11: Listing live → shared on WhatsApp communities (integration)

Screen 12: Lease agreement generation (if deal confirmed)

Journey 4: PG Seeker Finds and Books PG

Screen 1: "Looking for PG" → Select gender, location, budget

Screen 2: PG results with photos, star rating, price, distance to college

Screen 3: Open PG listing → photos gallery, room types, meal plan, rules, reviews

Screen 4: Real-time availability: "3 Single rooms available"

Screen 5: "Book a Visit" → instant time slot selection

Screen 6: Visit confirmation QR code generated

Screen 7: Visit → caretaker scans QR → check-in logged

Screen 8: Post-visit: "Book This Room" → pay ₹1,000 token

Screen 9: Move-in checklist: Photo-based room condition documentation

Screen 10: Monthly payment configured → reminder on specified date

Screen 11: Community features: Meet other residents, share concerns

Journey 5: Rent Payment Flow

Screen 1: "Pay Rent" → select saved property or add new

Screen 2: Enter amount → purpose: "Rent - January 2025"

Screen 3: Enter owner's UPI ID / bank account (verified)

Screen 4: Payment options: UPI (₹0 fee) / Card (2% fee shown)

Screen 5: Confirm → UPI app opens for payment

Screen 6: Success → receipt generated (PDF, downloadable)

Screen 7: Owner receives notification + PDF receipt

Screen 8: Rent history: All payments with dates, amounts, receipts

Screen 9: Auto-reminder configured for next month

SECTION 8 — DEAL CONFIRMATION & TRANSACTION LOGIC

8.1 Deal Confirmation Events (Trigger Matrix)

Event	Confidence Level	Action
Rental agreement signed via FC platform	100%	Auto-mark deal confirmed
First rent paid via FC platform (same property/owner)	95%	Auto-mark deal confirmed
RM manually confirms (Relax plan)	90%	Mark confirmed after both parties' verbal OK
Owner marks listing "Rented"	70%	Tentative confirmation, ask tenant to confirm
Tenant marks listing "Taken"	70%	Tentative confirmation, ask owner to confirm
Both parties confirm (mutual)	100%	Confirmed

8.2 Deal Record Schema

json

```
{
  "dealId": "FC-DEAL-2024-10892",
  "dealType": "rental",
  "listingId": "FC-2024-BLR-10234",
  "tenantId": "user_abc123",
  "ownerId": "user_xyz789",
  "rmId": "rm_001",
  "confirmedAt": "2024-01-25T14:00:00Z",
  "confirmationMethod": "agreement_signed",
  "rentAmount": 25000,
  "depositAmount": 75000,
  "leaseStartDate": "2024-02-01",
  "leaseEndDate": "2025-01-31",
  "leaseDurationMonths": 12,
  "paymentChannel": "upi",
  "agreementId": "AGR-2024-00981",
  "propertyCity": "Bengaluru",
  "propertyLocality": "Whitefield",
  "dealRevenue": {
    "agreementFee": 1499,
    "subscriptionRevenue": 1499,
    "rentPayMDR": 375
  },
  "status": "confirmed",
  "disputeFlag": false
}
```

8.3 Impact of Deal Confirmation

On Metrics:

- Conversion rate updated (views → contacts → visits → deal)
- RM performance updated
- Listing auto-deactivated (or marked "Rented - Waiting for next availability")

On Incentives:

- RM commission calculated and queued for payout
- Tenant referral credit awarded (if referred)
- Owner loyalty points credited

On Reporting:

- Deal logged in revenue dashboard

- City/locality deal heat map updated
- Avg. days-to-close metric updated

SECTION 9 — TECHNICAL ARCHITECTURE — ZERO-INR STACK

9.1 Architecture Philosophy

“Serverless-first. Pay only for traffic. Scale to billions. Cost: ₹0 until ₹10 Cr revenue.”

The entire platform is built on free-tier and usage-based services where fixed costs are ₹0:

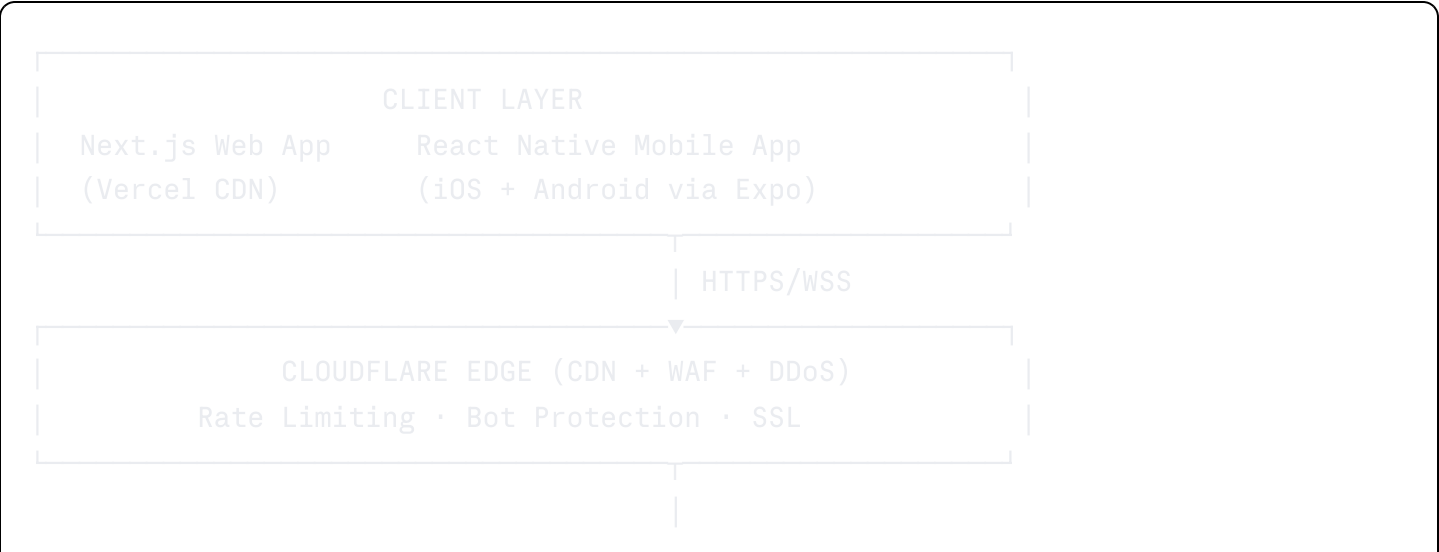
9.2 Zero-INR Infrastructure Stack

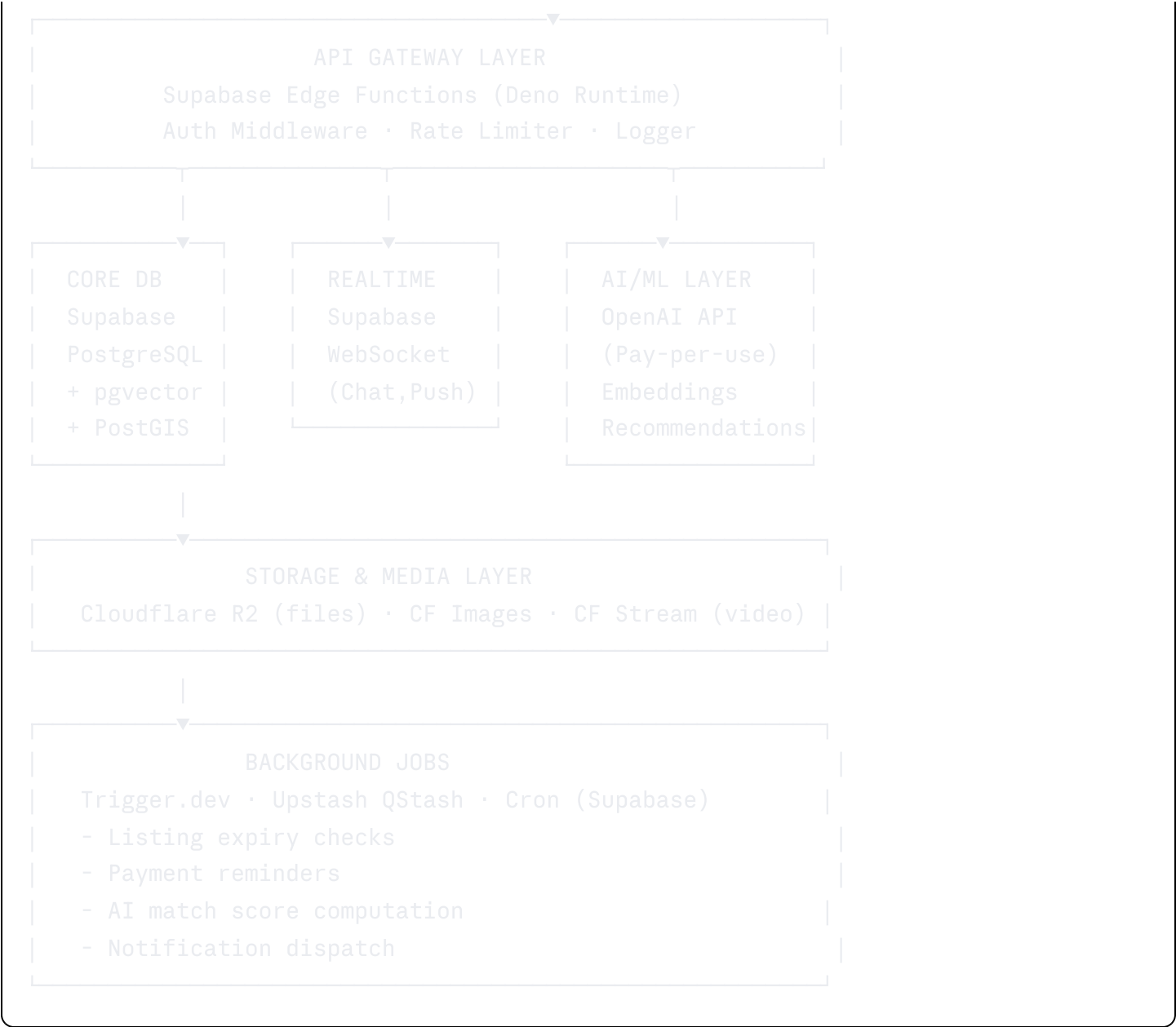
Layer	Service	Free Tier	Paid Trigger
Frontend	Next.js 14 on Vercel	100GB bandwidth/month	>100GB
Mobile	React Native (Expo)	Build free	Play Store: \$25 one-time
Backend API	Supabase Edge Functions	2M invocations/month	>2M
Database (Primary)	Supabase PostgreSQL	500MB storage, 2GB bandwidth	>500MB
Database (Scale)	PlanetScale (MySQL)	5GB, 1B row reads	>5GB
Auth	Supabase Auth	Unlimited	—
Realtime (Chat)	Supabase Realtime	Unlimited connections	—
File Storage	Cloudflare R2	10GB storage, 0 egress	>10GB (\$0.015/GB)
CDN	Cloudflare Free	Unlimited CDN	—
Image Processing	Cloudflare Images	100K transforms/month	>100K
Video	Cloudflare Stream	1000 min/month	>1000 min
Email	Resend	3,000 emails/month	>3,000 (\$20/mo)
SMS	MSG91 / Fast2SMS	Pay-per-SMS (~₹0.08/SMS)	Per use

Layer	Service	Free Tier	Paid Trigger
WhatsApp	Meta Business API	1,000 conversations/month free	>1,000
Maps	MapBox	50,000 loads/month	>50,000
Search (Vector)	Supabase pgvector	Included	—
Full-text Search	Supabase FTS	Included	—
Cache/Queue	Upstash Redis	10,000 commands/day	>10K
Background Jobs	Trigger.dev	50,000 runs/month	>50,000
Monitoring	Sentry Free	5,000 errors/month	>5,000
Analytics	PostHog Cloud	1M events/month	>1M
Logging	Logflare (Supabase)	5M logs/month	>5M
CI/CD	GitHub Actions	2,000 min/month	>2,000
Payments	Razorpay	0 fixed, 2%/transaction	Per transaction
KYC	DigiLocker API	Free (government)	—
E-sign	NSDL eSigning	Per sign (~₹15)	Per use
Calls	Agora	10,000 min/month free	>10,000 min

Total fixed monthly cost: ₹0 Variable cost model: Only pay when revenue flows in

9.3 System Architecture Diagram (Description)





9.4 Database Architecture

PostgreSQL Schema (Supabase)

sql

-- USERS TABLE

```
CREATE TABLE users (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  phone VARCHAR(13) UNIQUE NOT NULL,  
  email VARCHAR(255) UNIQUE,  
  name VARCHAR(100),  
  role user_role NOT NULL DEFAULT 'seeker',  
  kyc_status kyc_status_enum DEFAULT 'unverified',  
  kyc_level INTEGER DEFAULT 0, -- 0=none, 1=phone, 2=aadhaar, 3=document  
  aadhaar_hash VARCHAR(64), -- SHA-256 of masked Aadhaar (not stored raw)  
  pan_hash VARCHAR(64),  
  profile_photo_url TEXT,  
  city VARCHAR(100),  
  preferred_language language_enum DEFAULT 'english',  
  is_verified BOOLEAN DEFAULT FALSE,  
  is_blocked BOOLEAN DEFAULT FALSE,  
  rera_number VARCHAR(50),  
  created_at TIMESTAMPTZ DEFAULT NOW(),  
  updated_at TIMESTAMPTZ DEFAULT NOW(),  
  last_active_at TIMESTAMPTZ,  
  fcm_token TEXT, -- Firebase push token  
  metadata JSONB DEFAULT '{}'  
);
```

-- LISTINGS TABLE

```
CREATE TABLE listings (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  listing_code VARCHAR(30) UNIQUE NOT NULL, -- FC-2024-BLR-10234  
  owner_id UUID REFERENCES users(id),  
  property_type property_type_enum NOT NULL,  
  transaction_type transaction_type_enum NOT NULL,  
  status listing_status_enum DEFAULT 'draft',  
  title VARCHAR(200) NOT NULL,  
  description TEXT,  
  -- Location  
  address TEXT,  
  locality VARCHAR(100),  
  city VARCHAR(100) NOT NULL,  
  state VARCHAR(100) NOT NULL,  
  pincode VARCHAR(6),  
  latitude DECIMAL(10,8),  
  longitude DECIMAL(11,8),  
  geo_point GEOGRAPHY(POINT,4326), -- PostGIS
```

```
-- Residential fields
bhk INTEGER,
total_area_sqft INTEGER,
carpet_area_sqft INTEGER,
floor INTEGER,
total_floors INTEGER,
furnishing furnishing_enum,
facing facing_enum,
age_years INTEGER,
-- Pricing
rent_monthly INTEGER,
deposit_amount INTEGER,
maintenance_monthly INTEGER,
-- Status
available_from DATE,
is_negotiable BOOLEAN DEFAULT FALSE,
-- Agricultural fields
land_area_acres DECIMAL(10,3),
survey_number VARCHAR(50),
land_type land_type_enum,
soil_type soil_type_enum[],
water_source water_source_enum[],
lease_term_years INTEGER,
lease_price_per_acre_year INTEGER,
patta_number VARCHAR(50),
-- Metadata
photos TEXT[],
video_url TEXT,
virtual_tour_url TEXT,
amenities TEXT[],
preferences JSONB DEFAULT '{}',
ai_match_vector vector(1536), -- pgvector embedding
completeness_score INTEGER DEFAULT 0,
is_premium BOOLEAN DEFAULT FALSE,
is_verified BOOLEAN DEFAULT FALSE,
verification_notes TEXT,
verified_by UUID REFERENCES users(id),
verified_at TIMESTAMPTZ,
view_count INTEGER DEFAULT 0,
contact_count INTEGER DEFAULT 0,
shortlist_count INTEGER DEFAULT 0,
expires_at TIMESTAMPTZ,
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
```

```
);
```

```
-- DEALS TABLE
```

```
CREATE TABLE deals (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  deal_code VARCHAR(30) UNIQUE NOT NULL,  
  listing_id UUID REFERENCES listings(id),  
  tenant_id UUID REFERENCES users(id),  
  owner_id UUID REFERENCES users(id),  
  rm_id UUID REFERENCES users(id),  
  deal_type deal_type_enum NOT NULL,  
  status deal_status_enum DEFAULT 'initiated',  
  confirmation_method deal_confirmation_method_enum,  
  rent_amount INTEGER,  
  deposit_amount INTEGER,  
  lease_start_date DATE,  
  lease_end_date DATE,  
  agreement_id UUID REFERENCES agreements(id),  
  confirmed_at TIMESTAMPTZ,  
  revenue_snapshot JSONB DEFAULT '{}',  
  dispute_flag BOOLEAN DEFAULT FALSE,  
  notes TEXT,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

```
-- MESSAGES TABLE
```

```
CREATE TABLE messages (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  conversation_id UUID NOT NULL,  
  sender_id UUID REFERENCES users(id),  
  receiver_id UUID REFERENCES users(id),  
  listing_id UUID REFERENCES listings(id),  
  message_type message_type_enum DEFAULT 'text',  
  content TEXT,  
  media_url TEXT,  
  is_read BOOLEAN DEFAULT FALSE,  
  is_deleted_by_sender BOOLEAN DEFAULT FALSE,  
  created_at TIMESTAMPTZ DEFAULT NOW()  
);
```

```
-- PAYMENTS TABLE
```

```
CREATE TABLE payments (  
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  payment_code VARCHAR(30) UNIQUE NOT NULL,
```

```
payer_id UUID REFERENCES users(id),
payee_id UUID REFERENCES users(id),
payment_type payment_type_enum NOT NULL,
amount INTEGER NOT NULL, -- in paise
currency VARCHAR(3) DEFAULT 'INR',
status payment_status_enum DEFAULT 'initiated',
gateway VARCHAR(20),
gateway_order_id VARCHAR(100),
gateway_payment_id VARCHAR(100),
gateway_signature VARCHAR(255),
upi_transaction_id VARCHAR(100),
listing_id UUID REFERENCES listings(id),
deal_id UUID REFERENCES deals(id),
plan_id UUID REFERENCES subscription_plans(id),
receipt_url TEXT,
metadata JSONB DEFAULT '{}',
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

-- SUBSCRIPTIONS TABLE

```
CREATE TABLE subscriptions (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  user_id UUID REFERENCES users(id),
  plan_code VARCHAR(50) NOT NULL,
  plan_name VARCHAR(100) NOT NULL,
  user_role user_role NOT NULL,
  status sub_status_enum DEFAULT 'active',
  amount_paid INTEGER NOT NULL,
  starts_at TIMESTAMPTZ NOT NULL,
  expires_at TIMESTAMPTZ NOT NULL,
  payment_id UUID REFERENCES payments(id),
  rm_assigned UUID REFERENCES users(id),
  auto_renew BOOLEAN DEFAULT FALSE,
  created_at TIMESTAMPTZ DEFAULT NOW()
);
```

-- AGREEMENTS TABLE

```
CREATE TABLE agreements (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  agreement_code VARCHAR(30) UNIQUE NOT NULL,
  agreement_type agreement_type_enum NOT NULL,
  deal_id UUID REFERENCES deals(id),
  tenant_id UUID REFERENCES users(id),
```

```

owner_id UUID REFERENCES users(id),
listing_id UUID REFERENCES listings(id),
status agreement_status_enum DEFAULT 'draft',
stamp_duty INTEGER,
fc_fee INTEGER,
total_paid INTEGER,
document_url TEXT,
signed_document_url TEXT,
tenant_signed_at TIMESTAMPTZ,
owner_signed_at TIMESTAMPTZ,
delivery_address TEXT,
physical_delivery_opted BOOLEAN DEFAULT FALSE,
dispatched_at TIMESTAMPTZ,
tracking_number VARCHAR(50),
content JSONB NOT NULL, -- Full agreement terms
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);

```

-- REVIEWS TABLE

```

CREATE TABLE reviews (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  reviewer_id UUID REFERENCES users(id),
  reviewee_id UUID REFERENCES users(id),
  listing_id UUID REFERENCES listings(id),
  deal_id UUID REFERENCES deals(id),
  review_type review_type_enum NOT NULL,
  overall_rating DECIMAL(2,1) CHECK (overall_rating BETWEEN 1 AND 5),
  dimensions JSONB DEFAULT '{}', -- {cleanliness: 4.5, owner_behavior: 5, value: 4}
  comment TEXT,
  is_verified_deal BOOLEAN DEFAULT FALSE,
  upvotes INTEGER DEFAULT 0,
  is_published BOOLEAN DEFAULT TRUE,
  created_at TIMESTAMPTZ DEFAULT NOW()
);

```

-- SUPPORT TICKETS TABLE

```

CREATE TABLE support_tickets (
  id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
  ticket_code VARCHAR(20) UNIQUE NOT NULL,
  user_id UUID REFERENCES users(id),
  category ticket_category_enum NOT NULL,
  priority ticket_priority_enum DEFAULT 'medium',
  status ticket_status_enum DEFAULT 'open',

```

```
subject VARCHAR(200),
description TEXT,
attachments TEXT[],
assigned_to UUID REFERENCES users(id),
sla_deadline TIMESTAMPTZ,
resolved_at TIMESTAMPTZ,
resolution_notes TEXT,
satisfaction_rating INTEGER CHECK (satisfaction_rating BETWEEN 1 AND 5),
created_at TIMESTAMPTZ DEFAULT NOW(),
updated_at TIMESTAMPTZ DEFAULT NOW()
);
```

9.5 API Design

Base URL Structure

```
Production: https://api.farmerconnect.in/v1
Staging:    https://api-staging.farmerconnect.in/v1
```

Authentication

```
Type: JWT Bearer Token (Supabase Auth)
Header: Authorization: Bearer <supabase_jwt>
Refresh: /auth/refresh
Token expiry: 1 hour (access), 30 days (refresh)
```

Core API Endpoints

```
yaml
```

AUTH

POST	/auth/otp/send	# Send OTP to phone
POST	/auth/otp/verify	# Verify OTP → returns JWT
POST	/auth/refresh	# Refresh access token
DELETE	/auth/session	# Logout

USERS

GET	/users/me	# Get current user profile
PATCH	/users/me	# Update profile
POST	/users/kyc/initiate	# Start KYC process
GET	/users/kyc/status	# Get KYC status
POST	/users/kyc/upload	# Upload KYC documents

LISTINGS

GET	/listings	# Search listings (params: q, type, city, lat, lng, r)
POST	/listings	# Create listing
GET	/listings/:id	# Get listing detail
PATCH	/listings/:id	# Update listing
DELETE	/listings/:id	# Delete listing
POST	/listings/:id/photos	# Upload photos
GET	/listings/:id/contacts	# Get contact requests for this listing
POST	/listings/:id/shortlist	# Shortlist a listing
DELETE	/listings/:id/shortlist	# Remove from shortlist
POST	/listings/:id/report	# Report a listing
GET	/listings/my	# Get my listings (owner)
PATCH	/listings/:id/status	# Update status (active/paused/rented)

SEARCH

GET	/search/listings	# Full-text + vector search
GET	/search/suggest	# Autocomplete suggestions
GET	/search/nearby	# Nearby properties (lat, lng, radius)
GET	/search/agri	# Agricultural land search

CONTACTS

POST	/contacts	# Unlock contact (deducts credit or checks plan)
GET	/contacts/history	# Contact history

MESSAGES

GET	/conversations	# Get all conversations
POST	/conversations	# Start conversation
GET	/conversations/:id/messages	# Get messages
POST	/conversations/:id/messages	# Send message

DEALS

POST	/deals	# Initiate deal
GET	/deals/:id	# Get deal details
PATCH	/deals/:id/confirm	# Confirm deal
GET	/deals/my	# Get my deals

PAYMENTS

POST	/payments/order	# Create Razorpay order
POST	/payments/verify	# Verify payment signature
GET	/payments/history	# Payment history
GET	/payments/receipts/:id	# Download receipt PDF

SUBSCRIPTIONS

GET	/plans	# Get available plans
POST	/subscriptions	# Purchase subscription
GET	/subscriptions/current	# Get active subscription

AGREEMENTS

POST	/agreements	# Initiate agreement creation
GET	/agreements/:id	# Get agreement
POST	/agreements/:id/sign	# E-sign agreement
GET	/agreements/:id/download	# Download agreement PDF

RENT PAYMENTS

POST	/rent-payments/initiate	# Initiate rent payment to owner
GET	/rent-payments/history	# Rent payment history
GET	/rent-payments/schedule	# Upcoming rent schedule

SOCIETY

POST	/society	# Create society
GET	/society/:id	# Get society details
POST	/society/:id/residents	# Add resident
GET	/society/:id/notices	# Get notices
POST	/society/:id/notices	# Post notice
GET	/society/:id/visitors	# Get visitor log
POST	/society/:id/visitors	# Log visitor

SUPPORT

POST	/tickets	# Create support ticket
GET	/tickets/:id	# Get ticket
PATCH	/tickets/:id	# Update ticket

ANALYTICS (Internal)


```
GET    /analytics/listings    # Listing performance metrics
GET    /analytics/platform    # Platform-level metrics (admin)
```

SECTION 10 — SECURITY ARCHITECTURE — BANK-GRADE

10.1 Security Principles

- **Zero Trust:** Every request re-authenticated, no implicit trust
- **Defense in Depth:** Multiple layers of protection
- **Privacy by Design:** Minimal data collection, maximum encryption
- **DPDP Act 2023 Compliance:** India's data protection framework

10.2 Authentication & Authorization

Multi-Factor Authentication (MFA):

Layer 1: Phone OTP (mandatory for all users)

Layer 2: Aadhaar OTP (for KYC level 2+)

Layer 3: TOTP (Google Authenticator) for internal admins

Session Management:

- JWT with short expiry (1 hour access token)
- Refresh token rotation on each use
- Concurrent session limit: 5 devices
- Force logout: Admin can invalidate all sessions

Row-Level Security (RLS):

- Supabase RLS policies on every table
- Users can only read/write their own data
- Admin bypass via `service_role` key (backend-only, never exposed to client)
- Example policy:

```
CREATE POLICY "Users can only see own listings" ON listings
FOR SELECT USING (owner_id = auth.uid() OR status = 'active');
```

10.3 Data Encryption

At Rest:

- Supabase: AES-256 encryption at disk level
- Cloudflare R2: Server-side encryption (AES-256)

- Aadhaar numbers: Never stored raw; stored as SHA-256 hash
- PAN numbers: Masked (XXXXXX1234F format stored)
- Bank account numbers: Encrypted with AES-256-GCM before storage
- Credit card data: Never stored (Razorpay vaulted)

In Transit:

- TLS 1.3 minimum for all connections
- HSTS header enforced
- Certificate Transparency monitoring

Application Level:

- Chat messages: E2E encrypted (Signal Protocol via libsodium WASM)
- Call recordings: AES-256 at rest, deleted after 90 days
- Payment data: Never logged in application logs

10.4 API Security

Rate Limiting (Cloudflare + Upstash Redis):

- Auth endpoints: 5 requests/minute per IP
- Search: 60 requests/minute per user
- Contact unlock: 20/day per user (free tier)
- File upload: 100MB/day per user
- OTP send: 5 attempts/10 minutes per phone

Input Validation:

- All inputs validated on server (Zod schema validation)
- SQL injection prevention: Parameterized queries (Supabase client)
- XSS prevention: Content Security Policy headers
- SSRF prevention: Whitelist of allowed external URLs
- File upload: MIME type validation, virus scan (ClamAV on edge function)

Injection Prevention:

- All DB queries via Supabase SDK (never raw SQL from user input)
- Listing descriptions: Sanitized (DOMPurify server-side)
- Phone number regex validation: `/^[6-9]\d{9}$/`
- Pincode validation: `/^\d{6}$/`

API Key Management:

- No keys in frontend code (all in Vercel env vars)
- Razorpay: Server-side signature verification on every webhook
- Third-party keys rotated quarterly
- Secrets stored in Supabase Vault (encrypted KV)

10.5 Fraud Detection

Listing Fraud Signals:

- Duplicate photo detection (perceptual hash)
- Price anomaly: >40% below market → flag for review
- Phone number in description: Auto-redact + flag
- "Broker" keywords detected: Auto-flag
- New account + immediate listing: Extra verification required
- Multiple listings from same IP (different accounts): Alert

User Fraud Signals:

- Multiple accounts from same Aadhaar hash: Block + alert
- Mass contact unlock (>50/day): Suspicious → rate limit
- Fake review detection: ML model on review text patterns
- Payment fraud: Razorpay fraud detection integrated

Automated Responses:

- Risk score < 30: Auto-block
- Risk score 30-60: Manual review
- Risk score > 60: Auto-approve
- Repeated flags: Account suspension → support ticket for appeal

10.6 Compliance

DPDP Act 2023:

- Privacy policy: Plain language, vernacular versions
- Consent: Explicit, granular, withdrawable
- Data minimization: Only collect what's needed
- Right to erasure: Account deletion API
- Data retention: Defined per data type:
 - Active listing data: Retained while active + 6 months
 - Deal records: 7 years (legal requirement)
 - Chat messages: 90 days after conversation closes
 - Call recordings: 90 days
 - KYC documents: 5 years (PMLA requirement)
 - Payment records: 7 years (Income Tax)
 - Marketing data: Until consent withdrawn

IT Act 2000 Compliance:

- E-sign under Section 5 (legal validity)
- Intermediary guidelines (Rule 3(1)(b) content moderation)
- Grievance officer: Named, contact published (mandatory)

- CSAM policy: Zero tolerance, NCMEC reporting

RERA Compliance:

- Broker RERA number mandatory for broker listings
- Agent listings flagged if RERA expired
- State-specific RERA rules applied

RBI Payment Compliance:

- PCI DSS via Razorpay (no card data on FC servers)
- UPI compliance via licensed payment aggregators
- Settlement: T+2 for rent payments (per RBI guidelines)

SECTION 11 — UI/UX DESIGN SYSTEM

11.1 Design Philosophy

"Familiar but better" — The design should feel instantly familiar to Indian smartphone users (WhatsApp-like simplicity) but distinctly superior in quality and trustworthiness.

Core Design Tenets:

- Mobile-first (80% of traffic will be mobile)
- Vernacular-ready (RTL and Indic script support)
- Low-bandwidth friendly (compressed images, skeleton screens)
- Accessibility: WCAG 2.1 AA compliant
- One-thumb operation on 5.5" screens

11.2 Design Tokens

CSS

```
/* COLOR SYSTEM */
:root {
  /* Primary: Trust Green */
  --color-primary-50: #f0fdf4;
  --color-primary-100: #dcfce7;
  --color-primary-500: #22c55e;
  --color-primary-600: #16a34a;
  --color-primary-700: #15803d;
  --color-primary-900: #14532d;

  /* Secondary: Earth Brown */
  --color-secondary-500: #92400e;
  --color-secondary-600: #78350f;

  /* Accent: Saffron */
  --color-accent-500: #f59e0b;
  --color-accent-600: #d97706;

  /* Semantic */
  --color-success: #22c55e;
  --color-warning: #f59e0b;
  --color-error: #ef4444;
  --color-info: #3b82f6;

  /* Neutral */
  --color-gray-50: #f9fafb;
  --color-gray-100: #f3f4f6;
  --color-gray-500: #6b7280;
  --color-gray-900: #111827;

  /* Typography */
  --font-display: 'Mukta', 'Noto Sans Telugu', sans-serif;
  --font-body: 'Inter', 'Noto Sans', sans-serif;
  --font-mono: 'JetBrains Mono', monospace;

  /* Spacing */
  --space-1: 4px; --space-2: 8px; --space-3: 12px;
  --space-4: 16px; --space-6: 24px; --space-8: 32px;
  --space-12: 48px; --space-16: 64px;

  /* Radius */
  --radius-sm: 6px; --radius-md: 12px; --radius-lg: 20px;
  --radius-full: 9999px;
```

```
/* Shadow */
--shadow-sm: 0 1px 3px rgba(0,0,0,0.12);
--shadow-md: 0 4px 16px rgba(0,0,0,0.10);
--shadow-lg: 0 8px 32px rgba(0,0,0,0.12);
}
```

11.3 Component Library (Core Components)

Atoms:

- Button (Primary, Secondary, Ghost, Danger, Link) + Loading state
- Input (Text, Number, OTP, Phone, Search, Textarea)
- Badge (Verified, Premium, New, Sale, Lease)
- Avatar (with fallback initials)
- Icon (Phosphor icons library – 6,000+ icons)
- Spinner / Skeleton loader
- Toast notification

Molecules:

- PropertyCard (photo, BHK, rent, locality, verified badge, AI score)
- FilterChip (selectable filter tags)
- StarRating (read + write mode)
- PhoneInput (country code + number)
- PriceInput (₹ prefix, formatted)
- DocumentUpload (drag-drop + camera capture)
- MapPin (custom styled for different property types)
- ChatBubble (sent/received with status indicators)

Organisms:

- SearchBar (text + voice + near-me)
- ListingGrid (responsive, infinite scroll)
- FilterDrawer (bottom sheet on mobile)
- PropertyGallery (swipeable carousel + full-screen)
- OwnerContactCard (masked phone + chat + call)
- AgreementViewer (PDF embed with sign button)
- SocietyDashboard (stats + quick actions)

Templates:

- SearchResults Page
- Listing Detail Page
- Owner Dashboard
- Tenant Shortlist
- Chat Window

- Payment Page
- Agreement Page
- Profile / KYC Page

11.4 Navigation Architecture (Mobile)

Bottom Navigation (5 tabs):

-  Home / Search
-  Shortlist
-  Messages
-  My Activity (Listings / Visits / Deals)
-  Profile

Top navigation:

- City selector (Mumbai, Bengaluru, Hyderabad, etc.)
- Notification bell (badge count)
- Language toggle (vernacular)

Side Drawer (admin/RM users):

- Dashboard
- Assigned Tenants
- Analytics
- Support Queue
- Settings

11.5 Motion & Animation Guidelines

Principles:

- Duration: 150ms (micro) to 400ms (page transitions)
- Easing: ease-out for entries, ease-in for exits
- Reduce motion: Respect prefers-reduced-motion media query

Key Animations:

- Card enter: Fade up 8px (150ms ease-out)
- Filter apply: Slide in from right (250ms)
- Photo gallery: Native-feel swipe (spring physics)
- Map pins: Scale in with bounce (200ms)
- Deal confirmation: Confetti burst (300ms) + success card
- Loading: Skeleton shimmer (1.5s linear infinite)
- OTP input: Shake on error (300ms ease-out)
- Verified badge: Pulse on first render

SECTION 12 — ANALYTICS, REPORTING & INTELLIGENCE

12.1 Platform-Level Metrics (Admin Dashboard)

Real-Time Dashboard:

- Active listings count (by type, city)
- New registrations today
- Searches in last hour
- Payments processed today (amount + count)
- Deals confirmed today
- Active support tickets

Weekly/Monthly Reports:

- Funnel: Registrations → KYC → Listing/Search → Contact → Deal
- Revenue by stream (subscriptions, agreements, rent pay, ads)
- City-wise performance
- Plan conversion rates (free → paid)
- Churn rate by plan type
- NPS score trend
- RM performance matrix

12.2 Owner Analytics Dashboard

Per Listing Metrics:

- Total views (today / week / all time)
- Unique visitors
- Shortlist count
- Contact unlocks (who contacted: verified/unverified)
- Call connections count
- Visit schedules
- Conversion funnel: Views → Contacts → Visits → Deal

Listing Quality Score:

- Photo quality: /25
- Details completeness: /25
- Response rate: /25
- Review score: /25
- Total: /100 → displayed as "Property Score"

Insights:

- "Your listing gets 3x more views with 360° tour"
- "Properties with >8 photos get 60% more contacts"
- "Optimal rent for your locality: ₹22,000-₹25,000"

12.3 AI Intelligence Layer

Feature 1: Smart Matching Engine

Input: Seeker preference profile (stated + behavioral)

Output: Ranked listing recommendations

Algorithm: Collaborative filtering + content-based hybrid

Vector: pgvector cosine similarity on listing embeddings

Update frequency: Real-time on new listings + daily batch rerank

Feature 2: Fair Rent Predictor

Input: City, Locality, BHK, area, furnishing, amenities

Output: Fair rent range (min, median, max)

Model: Gradient boost on historical deals + active listings

Displayed: On listing creation for owners + on listing for seekers

Feature 3: Fraud Scorer

Input: User behavior, listing attributes, device signals

Output: Risk score (0-100)

Triggers: New listing, first contact, payment attempt

Action: Auto-approve / Review / Block

Feature 4: AI Listing Summary

Input: Listing details + photos

Output: 2-sentence human-readable summary ("Perfect for working couples – spacious east-facing flat with parking, 5 mins from metro.")

Model: GPT-4o-mini (pay-per-use, ~₹0.002/listing)

Feature 5: Vernacular Voice Search

Input: Voice in Telugu/Hindi/Kannada

Output: Structured search query

Model: OpenAI Whisper (free self-host) → intent extraction → search API

Feature 6: Deal Probability Predictor

Input: Contact history, visit count, days on market, seeker profile

Output: Deal probability % (shown to RM)

Action: RM prioritizes high-probability cases

SECTION 13 — THIRD-PARTY INTEGRATIONS

13.1 Integration Map

Integration	Provider	Purpose	Cost
Phone OTP	MSG91 / Fast2SMS	User authentication	₹0.08/OTP
WhatsApp Notifications	Meta WABA	Deal updates, reminders	1,000 free/month
Email	Resend	Receipts, agreements	3,000 free/month
Maps (Display)	MapBox	Property location display	50K free/month
Maps (Geocoding)	MapBox	Address → lat/lng	Included
Maps (Directions)	MapBox	Commute time estimation	Included
Satellite Imagery	MapBox Satellite	Agri land verification	Included
Payment Gateway	Razorpay	All payments	2% per txn
UPI	via Razorpay	Zero-cost for platform	—
KYC / Aadhaar	DigiLocker API	Identity verification	Free (Govt)
RERA Validation	RERA State APIs	Broker verification	Free (Govt)
Land Records	AP/TS Dharani API	Agri land verification	Free (Govt)
E-sign	NSDL eSigning	Agreement signing	₹15/sign
E-stamp	State Portal	Stamp paper generation	State fee
Crop Prices	Agmarknet API	MSP data for farmers	Free (Govt)
VoIP Calls	Agora	In-app calls	10K min free
Call Masking	Exotel	Number masking	Pay-per-call
AI / LLM	OpenAI	Summaries, embeddings	Pay-per-token
Push Notifications	Firebase FCM	Mobile push	Free
Analytics	PostHog	User analytics	1M events free
Error Tracking	Sentry	Error monitoring	5K errors free

Integration	Provider	Purpose	Cost
Virus Scan	ClamAV	File upload security	Free (open source)
Background Jobs	Trigger.dev	Scheduled tasks	50K runs free
Search (Advanced)	Typesense Cloud	Full-text search	1,000 docs free

13.2 Webhook Handlers

```
Razorpay Webhooks → /webhooks/razorpay:
- payment.captured → activate subscription / confirm payment
- payment.failed → notify user + retry prompt
- refund.created → notify user
- order.paid → trigger receipt generation job

MSG91 Webhooks → /webhooks/sms:
- delivery_report → log SMS delivery status

Agora Webhooks → /webhooks/agora:
- call_ended → log duration, trigger post-call NPS

DigiLocker Callbacks → /webhooks/digilocker:
- document_fetched → trigger KYC processing
```

SECTION 14 — NON-FUNCTIONAL REQUIREMENTS

14.1 Performance Targets

Metric	Target	Measurement
Search API response time (p50)	< 200ms	Supabase Dashboard
Search API response time (p99)	< 800ms	Sentry Performance
Page load (LCP)	< 2.5s on 4G	Google Lighthouse
Mobile TTI (Time to Interactive)	< 3.5s	Lighthouse
Photo upload time	< 3s (5 photos)	Custom metric
Chat message delivery	< 500ms	Supabase Realtime

Metric	Target	Measurement
Payment API response	< 1.5s	Razorpay + custom
Listing creation to live	< 2 hours	Internal SLA

14.2 Scalability Design

Horizontal Scaling:

- Supabase: Auto-scales (managed)
- Vercel: CDN auto-scales globally
- Cloudflare: 200+ PoPs globally
- Edge Functions: Serverless = infinite horizontal scale

Database Scaling:

- Connection pooling: Supabase PgBouncer
- Read replicas: Enabled at scale (\$25/month – only when needed)
- Caching strategy:
 - Hot listings cached in Upstash Redis (5 min TTL)
 - Search results cached by query hash (2 min TTL)
 - User profile cached (10 min TTL)
- Database indexes:
 - listings: (city, status, property_type, rent_monthly, geo_point GIST)
 - listings: full-text tsvector index
 - listings: pgvector IVFFLAT index for AI search
 - messages: (conversation_id, created_at)
 - payments: (payer_id, created_at)

Traffic Estimation:

- Yr 1: 10K DAU → Supabase free handles comfortably
- Yr 2: 1L DAU → Supabase Pro (\$25/month)
- Yr 3: 10L DAU → Supabase Team + read replica (\$200/month)
- Yr 5: 1Cr DAU → Supabase Enterprise (negotiated)

14.3 Availability & Reliability

Uptime SLA Target: 99.9% (8.7 hours downtime/year)

Dependency SLAs (provider guarantees):

- Supabase: 99.9%
- Vercel: 99.99%
- Cloudflare: 99.99%

- Razorpay: 99.99%

Failure Handling:

- Circuit breaker: Auto-fallback if AI service down (skip AI, show basic results)
- Payment fallback: If Razorpay down → switch to Cashfree
- Offline mode: PWA service worker caches last search results
- Graceful degradation: Core search + view works without auth

Backup Strategy:

- Supabase: Daily automated backups (PITR 7 days on Pro)
- Cloudflare R2: Versioned storage
- Configuration: Infrastructure as Code (Terraform) in GitHub

SECTION 15 — COMPLIANCE, LEGAL & KYC FRAMEWORK

15.1 KYC Levels & Access Rights

Level	Verification	Access Granted
0 — Unverified	None	Browse only
1 — Phone Verified	Phone OTP	Browse + shortlist + 3 contacts
2 — Identity Verified	Phone + Aadhaar	All seeker features
3 — Document Verified	+ Ownership docs / Patta	Create listings
4 — Premium Verified	+ Video KYC	Verified badge, higher trust score

15.2 Data Retention Schedule

Data Type	Retention	Reason
Active user data	While account active	Service delivery
Deleted account data	30 days (then anonymized)	DPDP erasure right
KYC documents	5 years from last transaction	PMLA compliance
Payment records	7 years	Income Tax Act

Data Type	Retention	Reason
Deal records	7 years	Legal evidence
Chat messages	90 days post-closure	Dispute resolution
Call recordings	90 days	Dispute resolution
Listing data	90 days post-expiry	Analytics
Server logs	30 days	Security

15.3 User Rights (DPDP Act 2023)

Right	Implementation
Right to access	"Download my data" in settings → JSON export
Right to correction	Edit profile, re-submit KYC
Right to erasure	"Delete my account" → 30-day grace → purge
Right to portability	Data export includes all listings, deals, messages
Right to grievance	Grievance officer contact in app footer
Right to nominate	Nominee can access account on death

15.4 Grievance Officer (Mandatory under IT Rules)

Name: [Company-designated officer]

Contact: grievance@farmerconnect.in

Phone: [Support number]

Address: [Registered office]

Response SLA: 48 hours (mandatory under Rule 3(1)(b))

Resolution SLA: 30 days

SECTION 16 — DEVELOPER EXECUTION PLAN

16.1 Team Structure

Role	Count	Tech Stack
Full-Stack Lead	1	Next.js, Supabase, TypeScript
Mobile Developer	1	React Native (Expo)
Backend/API	1	Supabase Edge Functions, Deno
UI/UX Designer	1	Figma
DevOps (minimal)	0.5	Vercel, GitHub Actions
QA Engineer	1	Playwright (automated testing)
AI/ML Engineer	0.5	Python, OpenAI, pgvector

16.2 Tech Stack Decisions

Frontend (Web):
Framework: Next.js 14 (App Router + RSC)
Language: TypeScript 5
Styling: Tailwind CSS 3.4
Components: shadcn/ui + custom
State: Zustand (client) + React Query (server state)
Forms: React Hook Form + Zod validation
Maps: react-map-gl (MapBox)
Animation: Framer Motion
Testing: Vitest + Playwright
Mobile (iOS + Android):
Framework: React Native (Expo SDK 51)
Navigation: Expo Router (file-based)
State: Zustand
Camera: expo-camera
Maps: react-native-maps + MapBox
Push: expo-notifications (FCM)
Biometric: expo-local-authentication
Backend:

Runtime: Deno (Supabase Edge Functions)
Language: TypeScript
ORM: Supabase JS client + raw SQL for complex queries
Validation: Zod
Auth: Supabase Auth
Queue: Trigger.dev + Upstash QStash
Caching: Upstash Redis

Infrastructure:

Database: Supabase (PostgreSQL 15 + pgvector + PostGIS)
File Storage: Cloudflare R2
CDN: Cloudflare
Hosting: Vercel (web) + EAS (mobile builds)
CI/CD: GitHub Actions
Monitoring: Sentry + PostHog + Logflare
Secrets: Vercel Env + Supabase Vault

16.3 Phase-by-Phase Implementation Plan

Phase 0: Foundation (Week 1-2)

Goals: Dev environment, project structure, CI/CD, auth

Tasks:

- [] Initialize Next.js 14 project (App Router)
- [] Initialize Expo React Native project
- [] Set up Supabase project (DB + Auth + Storage)
- [] Configure Cloudflare R2 bucket
- [] Set up Vercel deployment pipeline
- [] Configure GitHub Actions (test + build + deploy)
- [] Implement design token system (Tailwind config)
- [] Build Supabase Auth: phone OTP flow
- [] Implement JWT middleware for all routes
- [] Set up Sentry + PostHog analytics
- [] Database: Core schema migration (users, listings, deals)
- [] RLS policies for core tables
- [] Set up Resend email + MSG91 SMS connectors

Deliverable: Auth flow working end-to-end on web + mobile

Phase 1: Core Marketplace — Listings (Week 3–5)

Goals: Search, create listing, view listing

Tasks:

- [] Build listing creation form (multi-step, all property types)
- [] Photo upload → Cloudflare R2 (direct upload, signed URLs)
- [] Photo processing pipeline (resize, compress, enhance)
- [] Listing verification queue (internal tool)
- [] Full-text search (Supabase FTS with tsvector)
- [] Geospatial search (PostGIS ST_DWithin)
- [] Filter system (all 25+ filters)
- [] Listing detail page (gallery, info, contact CTA)
- [] Map view (MapBox integration)
- [] Shortlisting functionality
- [] Listing expiry + refresh
- [] My Listings dashboard (owner)

Deliverable: Owner can post listing; seeker can search and find it

Phase 2: Communication & Contacts (Week 6–7)

Goals: Chat, masked calls, contact unlock system

Tasks:

- [] Chat system (Supabase Realtime)
- [] E2E encryption implementation (libsodium)
- [] Contact unlock flow (credit deduction / plan check)
- [] Number-masked call initiation (Agora WebRTC)
- [] Conversation list view
- [] Message notifications (FCM push)
- [] Anti-spam ML model (basic: keyword + rate limit)
- [] Phone number redaction in messages

Deliverable: Seeker can contact owner via masked channel

Phase 3: Payments & Subscriptions (Week 8–9)

Goals: Pay for plans, rent payment, receipts

Tasks:

- [] Razorpay integration (orders + verification)

- [] Subscription plan purchase flow
- [] Plan activation webhook handler
- [] Invoice / receipt PDF generation (React-PDF)
- [] Rent payment flow (owner bank validation via penny drop)
- [] Rent payment history
- [] Monthly reminder scheduler (Trigger.dev cron)
- [] FC Credits wallet system
- [] Refund processing (Razorpay API)
- [] Payment dashboard (admin)

Deliverable: User can subscribe; tenant can pay rent online

Phase 4: KYC & Verification (Week 10-11)

Goals: Complete KYC, listing verification, verified badge

Tasks:

- [] Document upload (Aadhaar, PAN, property docs)
- [] OCR pipeline (Tesseract / AWS Textract free tier)
- [] Face liveness check integration (DigiLocker)
- [] RERA validation for brokers
- [] KYC agent review dashboard (internal)
- [] KYC status notification flow
- [] Verified badge display
- [] Listing auto-approval for verified owners
- [] Fraud score calculation

Deliverable: End-to-end KYC + listing verification working

Phase 5: Deal & Agreement (Week 12-13)

Goals: Deal confirmation, digital agreement, e-sign

Tasks:

- [] Deal initiation flow
- [] Agreement form (pre-filled from deal data)
- [] Agreement PDF generation (PDFKit)
- [] E-stamp integration (state portal)
- [] E-sign via Aadhaar OTP (NSDL eSigning API)
- [] Agreement status tracking (draft → signed → delivered)
- [] Agreement operations dashboard (internal)
- [] Physical delivery option integration
- [] Deal confirmation webhook (triggers all downstream events)

☐ Post-deal: review prompt, rent setup, listing deactivation

Deliverable: Complete deal lifecycle from contact to signed agreement

Phase 6: Agricultural Land Module (Week 14-16)

Goals: Agri land search, listing, verification, lease agreement

Tasks:

- ☐ Land listing form (AP/TS specific fields)
- ☐ Survey number + Dharani API integration
- ☐ Patta / EC document verification
- ☐ Satellite imagery display (MapBox Satellite)
- ☐ Agri-specific search + filters
- ☐ Land lease agreement template (state-specific)
- ☐ Agmarknet crop price integration
- ☐ Agri land verification dashboard (internal)
- ☐ Vernacular interface for Telugu farmers
- ☐ Voice search in Telugu

Deliverable: Farmer can list land; lessee can search, contact, lease

Phase 7: Society Management Module (Week 17-19)

Goals: Society admin features, resident app, maintenance billing

Tasks:

- ☐ Society creation + admin onboarding
- ☐ Resident invite + flat claiming flow
- ☐ Maintenance bill generation + collection
- ☐ Visitor management (QR code generation + scanning)
- ☐ Community noticeboard
- ☐ Maintenance ticket system
- ☐ Vendor directory
- ☐ Society analytics dashboard
- ☐ Emergency alert broadcast
- ☐ Push notifications for society events

Deliverable: Full society management suite working

Phase 8: AI Layer (Week 20-22)

Goals: Smart matching, fraud detection, price prediction

Tasks:

- [] pgvector setup + listing embeddings (OpenAI text-embedding-3-small)
- [] Semantic search implementation
- [] Seeker preference model (from behavior)
- [] AI match score computation + display
- [] Fair rent predictor (statistical model initially, ML later)
- [] Fraud detection ML model integration
- [] AI listing summary generation (GPT-4o-mini)
- [] Voice search (Whisper → intent extraction)
- [] Personalized home feed
- [] Deal probability predictor for RM

Deliverable: AI-powered search and fraud detection live

Phase 9: RM & Internal Tools (Week 23-24)

Goals: RM dashboard, support tools, admin panel

Tasks:

- [] RM assignment to Relax plan subscribers
- [] RM lead management dashboard
- [] Visit scheduling calendar
- [] Internal support ticket system (Chatwoot integration)
- [] Admin panel: user management, listing management
- [] Financial reconciliation dashboard
- [] Analytics and reporting (PostHog dashboards)
- [] Automated report emails to stakeholders

Deliverable: Full internal tooling for RM, support, finance

Phase 10: Polish, Testing & Launch (Week 25-26)

Goals: Production readiness, performance, security audit

Tasks:

- [] Full Playwright E2E test suite (50+ critical flows)
- [] Load testing (k6) to 10K concurrent users
- [] Security penetration test (external vendor or OWASP ZAP)

- [] Accessibility audit (aXe)
- [] Performance optimization (LCP, CLS, FID targets)
- [] SEO setup (sitemap, structured data, meta tags)
- [] App Store + Play Store submission
- [] CDN cache configuration optimization
- [] Monitoring alerts (Sentry + PostHog)
- [] DPDP compliance audit
- [] Soft launch: 500 beta users (Bengaluru + Hyderabad)
- [] Public launch

Deliverable: Production-ready platform, app store approved

SECTION 17 — AI PROMPT TEMPLATES FOR EACH MODULE

For Developer AI Assistants (Use These Exactly)

Prompt 1: Authentication Module

Build a complete phone OTP authentication system using:

- Supabase Auth (phone provider)
- Next.js 14 App Router
- TypeScript with Zod validation
- MSG91 as OTP provider (fallback to Supabase's built-in)
- Mobile: Expo React Native

Requirements:

- OTP sent to Indian phone numbers only (regex: /^[6-9]\d{9}\$/)
- Rate limit: 5 OTP attempts per 10 minutes per phone
- JWT stored in httpOnly cookie (web) / SecureStore (mobile)
- Auto-refresh before expiry
- Multi-device session management
- Logout from all devices option

Deliver:

- Supabase auth configuration
- Auth middleware for Next.js App Router
- Login page component (Tailwind styled)
- Mobile auth screen (React Native)
- Rate limiting logic (Upstash Redis)
- Test suite (Vitest)

Prompt 2: Listing Search Module

Build a high-performance property search system using:

- Supabase PostgreSQL (PostGIS for geospatial, pgvector for AI search, tsvector for full-text)
- Next.js 14 API routes (Supabase Edge Functions)
- TypeScript

Requirements:

- Text search on: title, description, locality, city, amenities
- Filter support: 25+ filters (BHK, rent range, furnishing, property type, etc.)
- Geospatial: radius search (ST_DWithin) from user lat/lng
- AI semantic search: pgvector cosine similarity on listing embeddings
- Sorting: relevance, rent (asc/desc), date, AI match score
- Pagination: cursor-based (not offset, for performance)
- Response: paginated listing cards (minimal data for list view)
- Caching: Upstash Redis (2-min TTL per unique query hash)
- Performance target: p95 < 500ms

Deliver:

- Database indexes (optimized for this query pattern)
- Search API endpoint with full filter support
- Cursor-based pagination implementation
- Redis caching layer
- Example queries with EXPLAIN ANALYZE results

Prompt 3: Real-Time Chat Module

Build a real-time encrypted chat system using:

- Supabase Realtime (PostgreSQL → WebSocket)
- libsodium (E2E encryption, WASM in browser)
- React (web) + React Native (mobile)

Requirements:

- One-to-one chats (seeker ↔ owner, per listing context)
- E2E encrypted (client-side encrypt before storing)
- Message types: text, photo, quick-reply
- Delivery status: sent → delivered → read
- Typing indicator (ephemeral, not stored)
- Auto-redact phone numbers from message content
- Block/report user
- 90-day message retention (auto-delete cron)
- Offline queue: send when reconnected

Deliver:

- Supabase Realtime subscription setup
- Encryption/decryption utilities (libsodium)
- Chat UI component (Tailwind)
- Photo sharing (Cloudflare R2 pre-signed upload)
- Phone number redaction middleware
- Message retention cron job (Trigger.dev)

Prompt 4: Payment Integration Module

Build a complete payment system using Razorpay:

- Subscription plan purchase
- Rent payment (tenant to owner)
- Agreement fee payment
- Receipt generation (React-PDF)

Requirements:

- Server-side order creation (never client-side)
- Razorpay signature verification on every webhook
- Idempotency keys to prevent duplicate processing
- UPI as default (no platform fee), card/NB with fee display
- Payment state machine: initiated → captured → settled / failed / refunded
- Automatic receipt PDF generation and email
- Webhook handlers: payment.captured, payment.failed, refund.created
- Admin dashboard: real-time payment metrics

Deliver:

- Razorpay integration (server: Next.js route handlers)
 - Payment state machine (Supabase RLS-protected table)
 - Webhook handler with signature verification
 - Receipt PDF template (React-PDF)
 - Razorpay checkout integration (client-side, secure)
 - Admin payments dashboard component
-

SECTION 18 — RISK REGISTER & EDGE CASE CATALOG

18.1 Business Risks

Risk	Probability	Impact	Mitigation
Broker infiltration (fake owner listings)	High	High	KYC + ownership doc verification; broker-detect AI
Fake listings (scam properties)	High	High	Multi-layer verification; real-time fraud score
NoBroker legal challenge	Medium	Medium	Operate differently — agri module is unique moat; legal counsel
Low conversion (free → paid)	Medium	High	RM-guided free trial; strong feature gating
Payment failures / UPI downtime	Low	Medium	Fallback gateway; auto-retry logic
Government data API downtime (Dharani, DigiLocker)	Medium	Low	Cache last known good response; queue retry
Data breach	Low	Very High	Encryption at rest/transit; penetration testing; cyber insurance

18.2 Technical Edge Cases

Edge Case 1: Same Aadhaar on multiple accounts Detection: Hash Aadhaar; check uniqueness before KYC approval Action: Block new account; send alert to admin; notify user to merge
Edge Case 2: Owner lists property already rented Detection: Check for active deals on same listing Action: Warn owner; auto-suspend after confirmation of old deal
Edge Case 3: Seeker pays rent to wrong person Detection: Bank validation before first payment; review period Action: Support ticket escalation; coordinate refund via Razorpay
Edge Case 4: Agreement signed but deal falls through Detection: Both parties mark "deal cancelled" within 7 days Action: Agreement voided (legal note); partial refund of FC fee

Edge Case 5: Agricultural land with disputed title

Detection: Litigation check via court records API (best-effort)

Action: "Dispute check pending" badge; warn lessee before deal

Edge Case 6: PG closes down mid-stay

Detection: Multiple resident complaints; owner not responding

Action: Residents get priority listings access; refund of advance

Edge Case 7: Society admin account hacked

Detection: Login from new device + bulk action triggered

Action: Auto-suspend bulk action; OTP re-verification; alert

Edge Case 8: Chat message with bank account request

Detection: Regex + ML model detects banking credential request

Action: Message hidden + warning shown; flag for review

Edge Case 9: Listing photos from stock image site

Detection: Reverse image search (Google Vision API or perceptual hash)

Action: Reject listing; ask for original photos with timestamp

Edge Case 10: User uploads Aadhaar with edited details

Detection: OCR + DigiLocker cross-check; document metadata analysis

Action: KYC rejected; account flagged; notify Aadhaar authority if pattern detected

18.3 Society Edge Cases

Edge Case 1: Multiple admins conflict

Resolution: Super-admin role hierarchy; majority vote for major decisions

Edge Case 2: Resident disputes maintenance charge

Resolution: Appeal to society committee; support ticket to FC

Edge Case 3: Owner of unit not resident (rented out)

Resolution: Both owner + tenant can be registered; owner approves tenant profile

Edge Case 4: Society RWA disputes FC's management role

Resolution: Clear ToS: FC is tool, not managing committee; data exportable

SECTION 19 — APPENDIX

Appendix A: Enum Definitions

sql

```
CREATE TYPE user_role AS ENUM ('seeker', 'owner', 'farmer', 'broker', 'pg_owner', 's
CREATE TYPE kyc_status_enum AS ENUM ('unverified', 'phone_verified', 'in_review', 'v
CREATE TYPE property_type_enum AS ENUM ('flat', 'house', 'villa', 'row_house', 'pg',
CREATE TYPE transaction_type_enum AS ENUM ('rent', 'sale', 'lease', 'pg', 'joint_far
CREATE TYPE listing_status_enum AS ENUM ('draft', 'under_review', 'active', 'paused'
CREATE TYPE furnishing_enum AS ENUM ('unfurnished', 'semi_furnished', 'fully_furnish
CREATE TYPE facing_enum AS ENUM ('north', 'south', 'east', 'west', 'north_east', 'no
CREATE TYPE deal_type_enum AS ENUM ('rental', 'sale', 'land_lease', 'pg_booking');
CREATE TYPE deal_status_enum AS ENUM ('initiated', 'in_progress', 'confirmed', 'canc
CREATE TYPE deal_confirmation_method_enum AS ENUM ('agreement_signed', 'rent_paid',
CREATE TYPE payment_type_enum AS ENUM ('subscription', 'rent_payment', 'agreement_fe
CREATE TYPE payment_status_enum AS ENUM ('initiated', 'captured', 'settled', 'failed
CREATE TYPE sub_status_enum AS ENUM ('active', 'expired', 'cancelled', 'paused');
CREATE TYPE agreement_type_enum AS ENUM ('rental_agreement', 'land_lease_agreement',
CREATE TYPE agreement_status_enum AS ENUM ('draft', 'pending_payment', 'paid', 'sent
CREATE TYPE land_type_enum AS ENUM ('agricultural', 'horticultural', 'plantation', '
CREATE TYPE soil_type_enum AS ENUM ('red', 'black', 'alluvial', 'sandy', 'laterite',
CREATE TYPE water_source_enum AS ENUM ('borewell', 'open_well', 'canal', 'drip_irrig
CREATE TYPE language_enum AS ENUM ('english', 'telugu', 'hindi', 'kannada', 'tamil',
CREATE TYPE message_type_enum AS ENUM ('text', 'photo', 'document', 'quick_reply', '
CREATE TYPE review_type_enum AS ENUM ('owner_review', 'listing_review', 'tenant_revi
CREATE TYPE ticket_category_enum AS ENUM ('listing_issue', 'payment_issue', 'agreeme
CREATE TYPE ticket_priority_enum AS ENUM ('low', 'medium', 'high', 'urgent');
CREATE TYPE ticket_status_enum AS ENUM ('open', 'assigned', 'in_progress', 'waiting_
```

Appendix B: Environment Variables

env

```
# Supabase
SUPABASE_URL=
SUPABASE_ANON_KEY=
SUPABASE_SERVICE_ROLE_KEY= # Never expose to client
```

```
# Cloudflare
CF_ACCOUNT_ID=
CF_R2_BUCKET_NAME=
CF_R2_ACCESS_KEY_ID=
CF_R2_SECRET_ACCESS_KEY=
CF_IMAGES_ACCOUNT_HASH=
```

```
# MapBox
NEXT_PUBLIC_MAPBOX_TOKEN=
```

```
# Razorpay
RAZORPAY_KEY_ID= # Public (client-safe)
RAZORPAY_KEY_SECRET= # Server only
RAZORPAY_WEBHOOK_SECRET= # Webhook verification
```

```
# MSG91
MSG91_AUTH_KEY=
MSG91_TEMPLATE_ID_OTP=
MSG91_TEMPLATE_ID_DEAL=
```

```
# Resend (Email)
RESEND_API_KEY=
```

```
# OpenAI
OPENAI_API_KEY=
```

```
# Upstash Redis
UPSTASH_REDIS_REST_URL=
UPSTASH_REDIS_REST_TOKEN=
```

```
# Trigger.dev
TRIGGER_API_KEY=
```

```
# Agora (Calls)
AGORA_APP_ID=
AGORA_APP_CERTIFICATE=
```

```
# Meta WhatsApp Business
```

```

WABA_TOKEN=
WABA_PHONE_NUMBER_ID=

# Sentry
SENTRY_DSN=
SENTRY_AUTH_TOKEN=

# PostHog
NEXT_PUBLIC_POSTHOG_KEY=
NEXT_PUBLIC_POSTHOG_HOST=

# App
NEXT_PUBLIC_APP_URL=https://farmerconnect.in
JWT_SECRET= # For custom JWT signing (if used alongside Supabase)
ENCRYPTION_KEY= # AES-256 for bank account encryption

```

Appendix C: Folder Structure

```

farmerconnect/
├─ apps/
│   └─ web/                                # Next.js 14
│       └─ app/                            # App Router
│           ├── (auth)/                    # Login, OTP
│           ├── (dashboard)/               # Owner/Admin dashboards
│           ├── search/                    # Search results
│           ├── listing/[id]/              # Listing detail
│           ├── messages/                  # Chat
│           ├── profile/                   # User profile, KYC
│           ├── pay/                       # Payment flows
│           ├── society/                   # Society management
│           └─ api/                         # API routes
│       └─ components/
│           ├── ui/                        # Design system atoms
│           ├── listing/                   # Listing-specific components
│           ├── search/                    # Search components
│           ├── chat/                      # Chat components
│           └─ layout/                     # Header, Footer, Navigation
│       └─ lib/
│           ├── supabase/                  # Supabase clients
│           ├── razorpay/                  # Payment utilities
│           ├── ai/                        # AI/ML utilities
│           ├── encryption/                # libsodium wrappers
│           └─ validators/                 # Zod schemas

```

```

|   |   └─ public/
|   └─ mobile/                # Expo React Native
|       └─ app/                # Expo Router
|       └─ components/
|       └─ lib/
└─ packages/
    └─ shared-types/          # TypeScript types (shared)
    └─ shared-validators/     # Zod schemas (shared)
    └─ shared-utils/          # Common utilities
└─ supabase/
    └─ migrations/            # DB migrations
    └─ functions/             # Edge Functions
    |   └─ search/            # Search Edge Function
    |   └─ webhooks/          # Payment webhooks
    |   └─ ai/                 # AI inference functions
    |   └─ cron/               # Scheduled jobs
    └─ seed/                   # Seed data
└─ docs/                       # Internal documentation
└─ .github/
    └─ workflows/             # CI/CD pipelines
└─ README.md

```

Appendix D: Launch Checklist

Pre-Launch (Week -2):

- [] Security penetration test completed
- [] DPDP compliance audit completed
- [] Grievance officer designated and contact published
- [] Privacy policy reviewed by lawyer (Indian law)
- [] Terms of Service reviewed by lawyer
- [] GST registration completed
- [] Razorpay KYC and settlement account configured
- [] MSG91 / Fast2SMS DLT registration (mandatory for transactional SMS)
- [] WhatsApp Business API approved
- [] Google Play Store: app submitted, approved
- [] Apple App Store: app submitted, approved
- [] Domain configured: farmerconnect.in + ssl
- [] Monitoring alerts configured (Sentry + PostHog)
- [] On-call rotation defined (who gets paged for P0 incidents)

Launch Day:

- [] Database: final seed data loaded
- [] Feature flags: disable unfinished features

- [] Smoke test: full user journey (register → post → search → contact → pay)
- [] Load test: 1,000 concurrent simulated users
- [] Social media: announcement posts scheduled
- [] Press release ready
- [] Customer support team briefed and on standby

Post-Launch (Week 1-4):

- [] Daily metric review (registrations, listings, contacts, payments)
- [] Weekly NPS survey to beta users
- [] Bug bash: fix all P0/P1 issues within 24 hours
- [] Referral program activated (Week 2)
- [] First content marketing push (SEO blog posts)
- [] Broker partnerships initiated
- [] Society admin outreach (target: 50 societies in Bengaluru/Hyderabad)

Appendix E: KPI Dashboard — Metrics That Matter

KPI	Definition	Target (Month 6)
Registered Users	Total signed-up users	1,00,000
MAU	Monthly Active Users	40,000
Active Listings	Live verified listings	10,000
Listing → Contact Rate	% of listed properties contacted	> 35%
Contact → Visit Rate	% of contacts leading to a visit	> 25%
Visit → Deal Rate	% of visits leading to a closed deal	> 20%
Overall Conversion	Listing → Closed Deal	> 1.75%
Avg. Days to Close	From listing to deal	< 21 days
Paid Conversion Rate	Free → Any Paid Plan	> 4%
ARPU	Avg Revenue Per User/Year	₹1,200
NPS	Net Promoter Score	> 45
Verified Listings %	% listings verified	> 90%
Fake Listing Rate	% listings found fraudulent	< 0.5%

KPI	Definition	Target (Month 6)
Support Ticket Rate	Tickets per 1000 users	< 15
App Store Rating	iOS + Android avg	> 4.3 ★

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